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Chapter 2

Accounting Information and Regression Analysis

2.1. Introduction

Accounting information, market information, and basic aggregated economic data are the basic inputs needed for financial analysis and planning; statistical methods, regression analysis, operation research programming techniques, and computer programming knowledge are important tools for achieving financial planning and forecasting. When performing financial analysis and planning, it is important to know how to use the appropriate tools in analyzing the relevant data.

The main purposes of this chapter are (i) to show how algebraic and statistical methods are used in cost–volume–profit (CVP) analysis and (ii) to demonstrate how regression methods can be used to analyze the dynamic adjustment process of financial ratios and obtain new insights into the use of financial ratios in financial analysis, planning, and forecasting. Recall that for financial management, the three major policies of the firm are investment, financing, and dividend policy. The basic concept of CVP analysis can be used in the areas of investment and financing, specifically for capital budget decision making and leverage analysis. Similarly, ratio analysis can be used to determine a firm's liquidity position, leverage position, the effectiveness of asset utilization, and profitability performance.

This chapter is organized as follows. Section 2.2 reviews four important financial statements: the balance sheet, the statement of earnings, the statement of equity, and the statement of cash flow. Section 2.3 discusses possible weaknesses of accounting information, and proposes possible methods to minimize these weaknesses. In Section 2.4, static ratio analysis is reviewed and dynamic financial ratio analysis is presented. Furthermore,

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both single-equation and simultaneous-equation approaches to dynamic financial analysis are explored. In Section 2.5, CVP analysis is extended from deterministic analysis to stochastic analysis. The concepts of statistical distributions are used to improve the robustness of CVP analysis. The relationship between accounting income and economic income is explored in Section 2.6. Finally, Section 2.7 summarizes the key concepts of the chapter. In addition, there are two appendixes to this chapter reviewing basic econometric methods. Appendix 2.A reviews basic concepts and methods of simple regression and multiple regressions, and Appendix 2.B discusses how to compile cash flow statement using the indirect method.

2.2. Financial Statements: A Brief Review

Corporate annual and quarterly reports generally contain four basic financial statements: balance sheet, statement of earnings, statement of retained earnings, and statement of changes in financial position. Using Johnson & Johnson (JNJ) annual consolidated financial statements as examples, we discuss the usefulness and problems associated with each of these statements in financial analysis and planning. Finally, the use of annual *versus* quarterly financial data is addressed.

2.2.1. *Balance Sheet*

The balance sheet describes a firm's financial position at one specific point in time. It is a static representation, such as a snapshot, of the firm's financial composition of assets and liabilities at one point in time. The balance sheet of JNJ, shown in Table 2.1, is broken down into two basic areas of classification — total assets (debit) and total liabilities and shareholders' equity (credit).

On the debit side, accounts are divided into six groups: current assets, marketable securities — non-current, property, plant and equipment (PP&E), intangible assets, deferred taxes on income, and other assets. Current assets represents short-term accounts, such as cash and cash equivalents, marketable securities and accounts receivable, inventories, deferred tax on income, and prepaid expense. It should be noted that deferred tax on income in this group is a current deferred tax and will be converted into income tax within 1 year.

Property encompasses all fixed or capital assets such as real estate, plant and equipment, special tools, and the allowance for depreciation and

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Table 2.1. Consolidated balance sheets of Johnson & Johnson Corporation and consolidated subsidiaries (2007–2012) (Dollars in Millions).

Assets	2007	2008	2009	2010	2011	2012
Current Assets						
Cash and Cash Equivalent ($)	7,770	10,768	15,810	19,355	24,542	14,911
Marketable Securities	1,545	2,041	3,615	8,303	7,719	6,178
Account Receivable	9,444	9,719	9,646	9,774	10,581	11,309
Inventory	5,110	5,052	5,180	5,378	6,285	7,495
Deferred Taxes on Income	2,609	3,430	2,793	2,224	2,556	3,139
Prepaid Expenses and Other Receivable	3,467	3,367	2,497	2,273	2,633	3,084
Total Current Assets	**29,945**	**34,377**	**39,541**	**47,307**	**54,316**	**46,116**
	—	—	—	—	—	—
Marketable Securities — Non-current	2	—	—	—	—	—
PP&E, Net	14,185	14,365	14,759	14,553	14,739	16,097
Intangible Assets, Net	28,763	27,695	31,185	32,010	34,276	51,176
Deferred Taxes on Income	4,889	5,841	5,507	5,096	6,540	4,541
Other Assets	3,170	2,634	3,690	3,942	3,773	3,417
Total Assets	**80,954**	**84,912**	**94,682**	**102,908**	**113,644**	**121,347**
Liabilities and Shareholder's Equity	—	—	—	—	—	—
Current Liabilities	—	—	—	—	—	—
Loans and Notes Payable	2,463	3,732	6,318	7,617	6,658	4,676
Accounts Payable	6,909	7,503	5,541	5,623	5,725	5,831
Accrued Liabilities	6,412	5,531	5,796	4,100	4,608	7,299
Accrued Rebates, Returns, and Promotion	2,318	2,237	2,028	2,512	2,637	2,969
Accrued Salaries, Wages, and Commissions	1,512	1,432	1,606	2,642	2,329	2,423
Taxes on Income	223	417	442	578	854	1,064
Total Current Liabilities	**19,837**	**20,852**	**21,731**	**23,072**	**22,811**	**24,262**
Long-term Debt	7,074	8,120	8,223	9,156	12,969	11,489
Deferred Tax liability	1,493	1,432	1,424	1,447	1,800	3,136
Employee Related Obligations	5,402	7,791	6,769	6,087	8,353	9,082
Other Liabilities	3,829	4,206	5,947	6,567	10,631	8,552
Shareowners' Equity	—	—	—	—	—	—
Preferred Stock-without Par Value	—	—	—	—	—	—
Common Stock-Par Value $1.00	3,120	3,120	3,120	3,120	3,120	3,120

(*Continued*)

Table 2.1. (*Continued*)

Assets	2007	2008	2009	2010	2011	2012
Net Receivable from Employee Stock Plan	—	—	—	—	—	—
Accumulated Other Comprehensive Income	−693	−4,955	−3,058	−3,531	−5,632	−5,810
Retained Earnings	55,280	63,379	70,306	77,773	81,251	85,992
Less: Common Stock Held in Treasury	14,388	19,033	19,780	20,783	21,659	18,476
Total Shareowners' Equity	**43,319**	**42,511**	**50,588**	**56,579**	**57,080**	**64,826**
Total Liabilities and Shareholders' Equity	**80,954**	**84,912**	**94,682**	**102,908**	**113,644**	**121,347**

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amortization. Intangible assets refer to the assets of research and development (R&D).

The credit side of the balance sheet in Table 2.1 is divided into current liabilities, long-term liabilities, and shareowner's equity. Under current liabilities, the following accounts are included: accounts, loans, and notes payable; accrued liabilities; accrued salaries and taxes on income. Long-term liabilities include various forms of long-term debt, deferred tax liability, employee-related obligations, and other liabilities. The stockholder's equity section of the balance sheet represents the net worth of the firm to its investors. For example, as of December 31, 2012, JNJ had $0 million preferred stock outstanding, $3,120 million in common stock outstanding, and $85,992 million retained earnings. Sometimes there are preferred stock and hybrid securities (e.g., convertible bond and convertible preferred stock) on the credit side of the balance sheet.

The balance sheet is useful because it depicts the firm's financing and investment policies. The use of comparative balance sheets, those that present several years' data, can be used to detect trends and possible future problems. JNJ has presented on its balance sheet information from six periods: December 31, 2007, December 31, 2008, December 31, 2009, December 31, 2010, December 31, 2011, and December 31, 2012. The balance sheet, however, is static and therefore should be analyzed with caution in financial analysis and planning.

2.2.2. *Statement of Earnings*

JNJ's statement of earnings is presented in Table 2.2 and describes the results of operations for a 12-month period ending December 31. The usual

Table 2.2. Consolidated statements of earnings of JNJ Corporation and subsidiaries (2007–2012) (dollars in millions).

(Dollars in Millions Except Per Share Figures)	2007	2008	2009	2010	2011	2012
Sales to Customers ($)	61,095	63,747	61,897	61,587	65,030	67,224
Cost of Products Sold	17,751	18,511	18,447	18,792	20,360	21,658
Gross Profit	43,344	45,236	43,450	42,795	44,670	45,566
Selling, Marketing and administrative expenses	20,451	21,490	19,801	19,424	20,969	20,869
Research Expense	7,680	7,577	6,986	6,844	7,548	7,665
Purchased in-process R&D	807	181	—	—	—	1,163
Interest Income	−452	−361	−90	−107	−91	−64
Interest Expense, net of portion capitalized	296	435	451	455	571	532
Other (income) expense, Net	1,279	−1,015	547	−768	2,743	1,626
	30,061	26,307	27,695	25,848	32,309	31,791
Earnings before Provision for Taxes on Income	13,283	16,929	15,755	16,947	12,361	13,775
Provision for Taxes on Income	2,707	3,980	3,489	3,613	2,689	3,261
Net Earnings	10,576	12,949	12,266	13,334	9,672	10,853
Basic Net Earnings per Share ($)	3.67	4.62	4.45	4.85	3.54	3.94
Diluted Net Earnings per Share ($)	3.63	4.57	4.40	4.78	3.49	3.86

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income-statement periods are annual, quarterly, and monthly. Johnson has chosen the annual approach. Both the annual and quarterly reports are used for external as well as internal reporting. The monthly statement is used primarily for internal purposes, such as the estimation of sales and profit targets, judgment of controls on expenses, and monitoring progress toward longer-term targets. The statement of earnings is more dynamic than the balance sheet, because it reflects changes for the period. It provides an analyst with an overview of a firm's operations and profitability on a gross, operating, and net income basis. JNJ's income includes sales, interest income, and other net income/expenses. Costs and expenses for JNJ include the cost of goods sold (selling, marketing, and administrative expenses), depreciation, depletion, and amortization. The difference between income and cost and expenses results in the company's Net Earnings. A comparative statement of earnings is very useful in financial analysis and planning

because it allows insight into the firm's operations, profitability, and financing decisions over time. For this reason JNJ presents the statement of earnings of six consecutive years: 2007, 2008, 2009, 2010, 2011, and 2012. Armed with this information, evaluating the firm's future is easier.

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2.2.3. *Statement of Equity*

JNJ's statements of equity are shown in Table 2.3. These are the earnings that a firm retains for reinvestment rather than paying them out to shareholders in the form of dividends. The statement of equity is easily understood if it is viewed as a bridge between the balance sheet and the statement of earnings. The statement of equity presents a summary of those categories that have an impact on the level of retained earnings: the net earnings and the dividends declared for preferred and common stock. It also represents a summary of the firm's dividend policy and shows how net income is allocated to dividends and reinvestment. JNJ's equity is one source of funds for investment, and this internal source of funds is very important to the firm. The balance sheet, the statement of earnings, and the statement of equity allow us to analyze important firm decisions on the capital structure, cost of capital, capital budgeting, and dividend policy of that firm.

2.2.4. *Statement of Cash Flows*

Another extremely important part of the annual and quarterly report is the statement of cash flows. This statement is very helpful in evaluating a firm's use of its funds and in determining how these funds were raised. Statements of cash flow for JNJ are shown in Table 2.4. These statements of cash flow are composed of three sections: cash flows from operating activities, cash flows from investing activities, and cash flows from financing activities. The statement of cash flows, whether developed on a cash or working-capital basis, summarizes long-term transactions that affect the firm's cash position. For JNJ, the sources of cash are essentially provided by operations. Application of these funds includes dividends paid to stockholders and expenditures for PPE, etc. Therefore, this statement reveals some important aspects of the firm's investment, financing, and dividend policies; making it an important tool for financial planning and analysis.

The cash flow statement shows how the net increase or decrease in cash has been reflected in the changing composition of current assets and current liabilities. It highlights changes in short-term financial policies. It helps

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Table 2.3. Consolidated statements of equity of JNJ Corporation and subsidiaries (2007–2012) (Dollars in Millions).

(Dollars in millions)	Total	Comprehensive Income	Retained Earnings	Accumulated Other Comprehensive Income	Common Stock Issued Amount	Treasury Stock Amount
Balance, December 30, 2007	**$43,319**		**55,280**	**(693)**	**3,120**	**(14,388)**
Net earnings	12,949	12,949	12,949	—	—	—
Cash dividends paid	(5,024)	—	(5,024)	—	—	—
Employee stock compensation and stock option plans	2,180	—	175	—	—	2,005
Conversion of subordinated debentures	—	—	(1)	—	—	1
Repurchase of common stock	(6,651)	—	—	—	—	(6,651)
Business combinations	—	—	—	—	—	—
Other comprehensive income, net of tax:	—	—	—	—	—	—
Currency translation adjustment	(2499)	(2499)	—	(,2499)	—	—
Unrealized gains/(losses) on securities	(59)	(59)	—	(59)	—	—
Pension liability adjustment	(1,870)	(1,870)	—	(1,870)	—	—
Gains on derivatives & hedges	166	166	—	166	—	—
Reclassification adjustment	—	(27)	—	—	—	—
Total comprehensive income	—	8,660	—	—	—	—
Note receivable from ESOP	—	—	—	—	—	—
Balance, December 28, 2008	**$42,511**	—	**63,379**	**(4,955)**	**3,120**	**19,033**

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Table 2.3. (*Continued*)

(Dollars in millions)	Total	Comprehensive Income	Retained Earnings	Accumulated Other Comprehensive Income	Common Stock Issued Amount	Treasury Stock Amount
Net earnings	12,266	12,266	12,266	—	—	—
Cash dividends paid	(5,327)	—	(5,327)	—	—	—
Employee stock compensation and stock option plans	1,402	—	25	—	—	1,377
Conversion of subordinated debentures	2	—	(4)	—	—	6
Repurchase of common stock	(2,130)	—	—	—	—	(2,130)
Business combinations (Other)	(33)	—	(33)	—	—	—
Other comprehensive income, net of tax:	—	—	—	—	—	—
Currency translation adjustment	1,363	1,363	—	1,363	—	—
Unrealized gains/(losses) on securities	(55)	(55)	—	(55)	—	—
Pension Liability adjustment	565	565	—	565	—	—
Gains/(losses) on derivatives & hedges	24	24	—	24	—	—
Reclassification adjustment	—	(15)	—	—	—	—
Total comprehensive income	—	14,163	—	—	—	—
Note receivable from ESOP	—	—	—	—	—	—
Balance, January 3, 2010	**50,588**	**—**	**70,306**	**(3,058)**	**3,120**	**(19,780)**

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Table 2.3. (*Continued*)

(Dollars in millions)	Total	Comprehensive Income	Retained Earnings	Accumulated Other Comprehensive Income	Common Stock Issued Amount	Treasury Stock Amount
Net earnings	13,334	13,334	13,334	—	—	—
Cash dividends paid	(5,804)	—	(5,804)	—	—	—
Employee stock compensation and stock option plans	1,730	—	(62)	—	—	1,792
Conversion of subordinated debentures	1	—	(1)	—	—	2
Repurchase of common stock	(2,797)	—	—	—	—	(2,797)
Other comprehensive income, net of tax:	—	—	—	—	—	—
Currency translation adjustment	(461)	(461)	—	(461)	—	—
Unrealized gains/(losses) on securities	54	54	—	54	—	—
Pension liability adjustment	(21)	(21)	—	(21)	—	—
Gains/(losses) on derivates & hedges	(45)	(45)	—	(45)	—	—
Reclassification adjustment	—	(9)	—	—	—	—
Total comprehensive income	—	12,861	—	—	—	—
Note receivable from ESOP	—	—	—	—	—	—
Balance, January 2, 2011	**$56,579**	**—**	**77,773**	**(3,531)**	**3,120**	**(20,783)**

(*Continued*)

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Table 2.3. (*Continued*)

(Dollars in millions)	Total	Comprehensive Income	Retained Earnings	Accumulated Other Comprehensive Income	Common Stock Issued Amount	Treasury Stock Amount
Net earnings	9,672	9,672	9,672	—	—	—
Cash dividends paid	(6,156)	—	(6,156)	—	—	—
Employee stock compensation and stock option plans	1,760	—	111	—	—	1,649
Conversion of subordinated debentures	—	—	—	—	—	—
Repurchase of common stock	(2,525)	—	—	—	—	(2,525)
Other	(149)	—	(149)	—	—	—
Other comprehensive income, net of tax:	—	—	—	—	—	—
Currency translation adjustment	(557)	(557)	—	(557)	—	—
Unrealized gains on securities	424	424	—	424	—	—
Pension liability adjustment	(1,700)	(1,700)	—	(1,700)	—	—
Losses on derivatives & hedges	(268)	(268)	—	(268)	—	—
Reclassification adjustment	—	—	—	—	—	—
Total comprehensive income	—	7,571	—	—	—	—
Note receivable from ESOP	—	—	—	—	—	—
Balance, January 1, 2012	**$57,080**	**—**	**81,251**	**(5,632)**	**3,120**	**(21,659)**

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Table 2.3. (*Continued*)

(Dollars in millions)	Total	Comprehensive Income	Retained Earnings	Accumulated Other Comprehensive Income	Common Stock Issued Amount	Treasury Stock Amount
Net earnings	10,853	10,853	10,853	—	—	—
Cash dividends paid	(6,614)	—	(6,614)	—	—	—
Employee stock compensation and stock option plans	3,269	—	19	—	—	3,250
Conversion of subordinated debentures	—	—	—	—	—	—
Repurchase of common stock	(12,919)	—	—	—	—	(12,919)
Other comprehensive income, net of tax:	—	—	—	—	—	—
Currency translation adjustment	1,230	1,230	—	1,230	—	—
Unrealized gains on securities	−253	−253	—	−253	—	—
Pension liability adjustment	−1,331	−1,331	—	−1,331	—	—
Gains on derivatives & hedges	176	176	—	176	—	—
Reclassification adjustment	—	—	—	—	—	—
Total comprehensive income	—	10,675	—	—	—	—
Note receivable from ESOP	—	—	—	—	—	—
Balance, December 30, 2012	**$64,826**	**—**	**85,992**	**(5,810)**	**3,120**	**(18,476)**

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Table 2.4. Consolidated Statements of Cash Flow of JNJ Corporation and Subsidiaries (Dollars in Millions).

(Dollars in Millions)	2007	2008	2009	2010	2011	2012
Cash flows from operating activities						
Net earnings	10,576	12,949	12,266	13,334	9,672	10,514
Adjustments to reconcile net earnings to cash flows:	—	—	—	—	—	—
Depreciation and amortization of property and intangibles	2,777	2,832	2,774	2,939	3,158	3,666
Purchased in-process R&D	807	181	—	—	—	—
Deferred tax provision	−1,762	22	−436	356	−836	−39
Accounts receivable allowances	22	86	58	12	32	92
Changes in assets and liabilities, net of effects from acquisitions:	—	—	—	—	—	—
Increase in accounts receivable	−416	−736	453	−207	−915	−9
(Increase)/decrease in inventories	14	−101	95	−196	−715	−1
(Decrease)/increase in accounts payable and accrued liabilities	2,642	−272	−507	20	493	2,768
Decrease/(increase) in other current and non-current assets	−1,578	−1,600	1,209	−574	−1,625	−2,172
Increase in other current and non-current liabilities	564	984	31	87	4,413	−2,555
Net cash flows from operating activities	13,646	14,345	15,943	16,385	14,298	16,385
Cash flows from investing activities	—	—	—	—	—	—
Additions to PP&E	−2,942	−3,066	−2,365	−2,384	−2,893	−2,934
Proceeds from the disposal of assets	457	785	154	524	1,342	1,509
Acquisitions, net of cash acquired	−1,388	−1,214	−2,470	−1,296	−2,797	−4,486
Purchases of investments	−9,659	−3,668	−10,040	−15,788	−29,882	−13,434
Sales of investments	7,988	3,059	7,232	11,101	30,396	14,797
Other (primarily intangibles)	−368	−83	−109	−38	−778	38
Net cash used by investing activities	−5,912	−4,187	−7,598	−7,854	−4,612	−4,510

(*Continued*)

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Table 2.4. (*Continued*)

(Dollars in Millions)	2007	2008	2009	2010	2011	2012
Cash flows from financing activities	—	—	—	—	—	—
Dividends to shareholders	−4,670	−5,024	−5,327	−5,804	−6,156	−6,614
Repurchase of common stock	−5,607	−6,651	−2,130	−2,797	−2,525	−12,919
Proceeds from short-term debt	19,626	8,430	9,484	7,874	9,729	3,268
Retirement of short-term debt	−21,691	−7,319	−6,791	−6,565	−11,200	−6,175
Proceeds from long-term debt	5,100	1,638	9	1,118	4,470	45
Retirement of long-term debt	−18	−24	−219	−32	−16	−804
Proceeds from the exercise of stock options	1,562	1,486	882	1,226	1,246	2,720
Other	—	—	—	—	—	−83
Net cash used by financing activities	−5,698	−7,464	−4,092	−4,980	−4,452	−20,562
Effect of exchange rate changes on cash and cash equivalents	275	−323	161	−6	−47	45
Increase/ (Decrease) in cash and cash equivalents	3,687	2,998	5,042	3,545	5,187	−9,631
Cash and cash equivalents, beginning of year	4,083	7,770	10,768	15,810	19,355	24,542
Cash and cash equivalents, end of year	7,770	10,768	15,810	19,355	24,542	14,911
Supplemental cash flow data	—	—	—	—	—	—
Cash paid during the year for:	—	—	—	—	—	—
Interest	314	525	533	491	576	616
Income taxes	4,099	4,068	2,368	2,442	2,970	2,507
Supplemental schedule of non-cash investing and financing activities	—	—	—	—	—	—
Treasury stock issued for employee compensation and stock option plans, net of cash proceeds	738	593	541	673	433	615
Conversion of debt	9	—	2	1	1	—
Acquisitions	—	—	—	—	—	—
Fair value of assets acquired	1,620	1,328	3,345	1,321	3,025	19,025
Fair value of liabilities assumed	−232	−114	−8,75	−52	−228	−1,204
Net cash paid for acquisitions	1,388	1,214	2,470	1,269	2,797	4,486

answer questions such as: Has the firm been building up its liquidity assets or is it becoming less liquid?

The statement of cash flow can be used to help resolve differences between finance and accounting theories. There is value for the analyst in viewing the statement of cash flow over time, especially in detecting trends that could lead to technical or legal bankruptcy in the future. Collectively, the balance sheet, the statement of retained earnings, the statement of equity, and the statement of cash flow present a fairly clear picture of the firm's historical and current position. The procedure for compiling Table 2.4 can be found in Appendix 2.B.

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2.2.5. *Interrelationship among Four Financial Statements*

It should be noted that the balance sheet, statement of earnings, statement of equity, and statement of cash flow are interrelated. These relationships are briefly described as follows:

(1) Retained earnings calculated from the statement of equity for the current period should be used to replace the retained earnings item in the balance sheet of the previous period. Therefore, the statement of equity is regarded as a bridge between the balance sheet and the statement of earnings.
(2) We need the information from the balance sheet, the statement of earnings, and the statement of equity to compile the statement of cash flow.
(3) Cash and cash equivalent item can be found in the statement of cash flow. In other words, the statement of cash flow describes how the cash and cash equivalent changed during the period. It is known that the first item of the balance sheet is cash and cash equivalent.

2.2.6. *Annual versus Quarterly Financial Data*

Both annual and quarterly financial data are important to financial analysts; which one is the most important depends on the time horizon of the analysis. Depending upon pattern changes in the historical data, either annual or quarterly data could prove to be more useful. As Gentry and Lee discuss, understanding the implications of using quarterly data versus annual data is important for proper financial analysis and planning.

Quarterly data has three components: trend-cycle, seasonal, and irregular or random components. It contains important information about

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seasonal fluctuations that "reflects an intra-year pattern of variation which is repeated constantly or in evolving fashion from year to year."[1] Quarterly data has the disadvantage of having a large irregular, or random, component that introduces noise into analysis.

Annual data has both the trend-cycle component and the irregular component, but it does not have the seasonal component. The irregular component is much smaller in annual data than in quarterly data. While it may seem that annual data would be more useful for long-term financial planning and analysis, seasonal data reveals important permanent patterns that underlie the short-term series in financial analysis and planning. In other words, quarterly data can be used for intermediate-term financial planning to improve financial management.

Use of either quarterly or annual data has a consistent impact on the mean-square error of regression forecasting (see Appendix 2.A), which is composed of variance and bias. Changing from annual to quarterly data will generally reduce variance while increasing bias. Any difference in regression results, due to the use of different data, must be analyzed in light of the historical patterns of fluctuation in the original time-series data.

2.3. Critique of Accounting Information

2.3.1. *Criticism*

At first glance, accounting information seems to be heavily audited and regulated, automatically determining what numbers are presented. However, careful analysis makes it apparent that accountants work with a fairly broad framework of rules that increase the distance between accounting and financial valuation. This leeway in accounting rules also tends to make accounting information more random. In addition, Hong (1977) shows that the selection of "last in, first out" (LIFO) or "first in, last out" (FIFO) methods for tax and depreciation based upon historic cost generally introduces a bias in a firm's market-value determination. This combination of discrepancy, bias, and randomness means that accounting information does not represent the "true" information. As a result, both time-series and cross-sectional comparisons of accounting information are difficult to analyze.

A major problem with the use of accounting information rises due to errors made in classifying transactions into individual accounts. There are several types of classification errors.

[1] *Ibid.*

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One classification error occurs when a bookkeeper enters an item into the wrong account. This is dealt with by modern auditing through the use of sampling techniques, where the auditor certifies that the probability of a material error, an error that would alter a manager's or investor's decision, is within an acceptable limit.

The difference between accountancy and finance theory is another case of classification error. An accountant defines income as the change in shareholder's wealth due to operations of the firm; this includes the use of accruals in wealth determination. The finance discipline defines a firm's income as cash income, or the difference between cash revenues and cash expenses (those payments made to generate current revenue). Due to the accruals used in accounting, accounting income is numerically different from cash income because of the difference in timing.

Another problem with accounting information derives from depreciation costs. There are various accepted ways to spread the cost of an asset over its useful life.[2] The choice of a depreciation method can cause a wide variation in net income.[3] A straight-line method will reduce income less than an accelerated method in the first years of depreciation. In the later years, accelerated depreciation will reduce income less than a straight-line method.

The use of historical costs for pricing an asset acquisition also causes problems in using accounting information. Such reliance on historical cost is particularly troublesome in times of high inflation, because historical cost values are no longer representative of the underlying values of the assets and liabilities of the firm. The accounting profession is attempting to deal with this problem through the use of supplementary disclosure of selected financial statement items, as required by Financial Accounting Standard

[2]Depreciation is a procedure that allocates the acquisition costs of the asset to subsequent periods of time on a systematic and rational basis. There are several widely used methods of allocating the acquisition costs: straight-line, declining balance, and sum-of-the-years digits.

[3]There is a limit on the ability of the firm to manipulate its financial statements by repeatedly changing accounting methods, for depreciation, inventory valuation, or any other of the numerous decisions permitted by generally accepted accounting principles (GAAP). The limitation on constant changes in accounting methods is provided by several sources. Accounting Principle Board 20 requires that any changes must be justified on the basis of fair representation and the change and the justification must be disclosed.

The American Institute of Certified Public Accountants, through the Auditing Standards Board, requires that the auditor certify that accounting principles have been consistently observed in the current period in relation to the preceding periods [AU 150.02].

Board 33, for large, publicly traded firms.[4] Accountants are also developing replacement cost and other inflation-adjusted accounting procedures.

2.3.2. *Methods for Improvement*

Three possible methods for improving the representativeness or accuracy of accounting information in financial analysis and planning is the use of alternative information, statistical tools, and finance and economic theories.

Use of Alternative Information

Of the many types of alternative information that is used to improve accounting data, the most practical and consistent type is market information. Stock prices and replacement costs can be used to adjust reported accounting earnings. According to the theory of efficient capital markets, the market price of a security represents the market's estimate of the value of that security. Furthermore, the market value (or the "intrinsic" value) of a common stock represents the firm's "true" earning potential perceived by investors. An example of the use of market information to complement accounting information is the use of the option-pricing model in capital budgeting under uncertainty in Chapter 10.

Statistical Adjustments Another method of improving accounting information is the use of statistical tools. By using a time-series decomposition technique suggested by Gentry and Lee (1983), quarterly earnings can be divided into three components: trend-cycle, seasonal, and irregular components. This decomposition procedure can be used to remove some undesirable noise associated with accounting numbers. Therefore, this statistically adjusted accounting earnings data can be used to improve its usefulness in determining a firm's intrinsic value.

For long-term financial planning and analysis, the trend-cycle component is the major source of information. The seasonal and irregular components introduce noise that clouds the analysis, and this noise can be removed. For short-term (or intermediate-term) planning and analysis, the seasonal component also produces valuable information. Thus, both trend-cycle and seasonal components should be used in working-capital management. Note that the source(s) of noise can also be eliminated by moving average or other statistical methods.

[4]Financial Accounting Standards Board Statement #33, Stamford CT, June 1974.

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Application of Finance and Economic Theories The third method of improving accounting information is the use of finance and economic theories. For example, there is the Modigliani and Miller valuation theory, the capital-asset pricing theory, and the option-pricing theory, which will be discussed in Chapters 4, 9, and 17. By applying these theories, one can adjust accounting income to obtain a better picture of income measurement, i.e., finance income (cash flow). Also, the use of finance theory combined with market and other information gives an analyst another estimate of income measurement. In addition, the various earnings estimates can shed additional light on the firm's value determination. To do these kinds of empirical tests, Lee and Zumwalt (1981) use a multiple regression model to investigate the association between six alternative accounting profitability measures and the security rate-of-return determination. Their empirical results suggest that accounting profitability information is an important extra-market information component of asset pricing. In other words, it is shown that different accounting profitability measures should be used by security analysts and investors to determine the equity rates-of-return for different industries.

2.4. Static Ratio Analysis and its Extension

In order to make use of financial statements, an analyst needs some form of measurement for analysis. Frequently, ratios are used to relate one piece of financial data to another. The ratio puts the two pieces of data on an equivalent base, which increases the usefulness of the data. For example, net income as an absolute number is meaningless to compare across firms of different sizes. However, if one creates a net profitability ratio (NI/Sales), comparisons are easier to make. Analysis of a series of ratios will give us a clear picture of a firm's financial condition and performance.

Analysis of ratios can take one of two forms. First, the analyst can compare the ratios of one firm with those of similar firms or with industry averages at a specific point in time. This is a type of cross-sectional analysis technique, that may indicate the relative financial condition and performance of a firm. One must be careful, however, to analyze the ratios while keeping in mind the inherent differences between a firm's production functions and its operations. Also, the analyst should avoid using "rules of thumb" across industries because the composition of industries and individual firms varies considerably. Furthermore, inconsistency in a firm's accounting procedures can cause accounting data to show substantial

differences between firms, which can hinder ratio comparability. This variation in accounting procedures can also lead to problems in determining the "target ratio" (to be discussed later).

The second method of ratio comparison involves the comparison of a firm's present ratio with their past and expected ratios. This form of time-series analysis will indicate whether the firm's financial condition has improved or deteriorated. Both types of ratio analysis can take one of the two following forms: static determination and its analysis, or dynamic adjustment and its analysis.

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2.4.1. *Static Determination of Financial Ratios*

The static determination of financial ratios involves the calculation and analysis of ratios over a number of periods for one company, or the analysis of differences in ratios among individual firms in one industry. An analyst must be careful of extreme values in either direction, because of the interrelationships between ratios. For instance, a very high liquidity ratio is costly to maintain, causing profitability ratios to be lower than they need to be. Furthermore, ratios must be interpreted in relation to the raw data from which they are calculated, particularly for ratios that sum accounts in order to arrive at the necessary data for the calculation. Even though this analysis must be performed with extreme caution, it can yield important conclusions in the analysis for a particular company. Table 2.5 presents ratio data for the domestic pharmaceutical industry.

Short-term Solvency, or Liquidity Ratios

Liquidity ratios are calculated from information on the balance sheet; they measure the relative strength of a firm's financial position. Crudely interpreted, these are coverage ratios that indicate the firm's ability to meet short-term obligations. The current ratio (CR) (ratio 1 in Table 2.5) is the most popular of the liquidity ratios because it is easy to calculate and it has intuitive appeal. It is also the most broadly defined liquidity ratio, as it does not take into account the differences in relative liquidity among the individual components of current assets. A more specifically defined liquidity ratio is the quick or acid-test ratio (ratio 2), which excludes the least liquid portion of current assets, inventories. Cash ratio (ratio 3) is the ratio of the company's total cash and cash equivalents to its current liabilities. It is most often used as a measure of company liquidity. A strong cash ratio is

Table 2.5. Company ratios period 2011–2012.

Ratio Classification	Formula	JNJ 2011	JNJ 2012	Industry 2008	Industry 2009
Short-term Solvency, or Liquidity Ratios					
(1) CR	$\frac{\text{Current asset}}{\text{Current liabilities}}$	2.38	1.9	—	—
(2) Quick Ratio*	$\frac{\text{Cash+MS+receivables}}{\text{Current liabilities}}$	1.88	1.34	—	—
(3) Cash Ratio*	$\frac{\text{Cash+MS}}{\text{Current liabilities}}$	1.41	0.87	—	—
(4) Networking-capital to total asset	$\frac{\text{Net working-capital}}{\text{Total asset}}$	0.28	0.18	—	—
Long-term solvency, or Financial Leverage Ratios					
(5) Debt-to-Asset	$\frac{\text{Total debt}}{\text{Total asset}}$	0.48	0.45	—	—
(6) Debt-to-Equity	$\frac{\text{Total debt}}{\text{Total equity}}$	0.96	0.84	—	—
(7) Equity Multiplier	$\frac{\text{Total asset}}{\text{Total equity}}$	2.00	1.87	—	—
(8) Times Interest Paid	$\frac{\text{EBIT}}{\text{Interest expenses}}$	22.65	26.89	—	—
(9) Long-term Debt Ratio	$\frac{\text{Long-term debt}}{\text{Long-term debt+Total equity}}$	0.19	0.15	—	—
(10) Cash Coverage Ratio	$\frac{\text{EBIT+Depreciation}}{\text{Interest expenses}}$	28.18	33.78	—	—
Asset Management, or Turnover (Activity) Ratios					
(11) Day's Sales in Receivables (Average Collection Period)	$\frac{\text{Account receivable}}{\text{Sales}/365}$	59.39	61.40	—	—
(12) Receivable Turnover	$\frac{\text{Sales}}{\text{Accounts receivable}}$	6.15	5.94	—	—
(13) Day's Sales in Inventory	$\frac{\text{Inventory}}{\text{Cost of goods sold}/365}$	112.67	126.31	—	—
(14) Inventory-Turnover	$\frac{\text{Cost of goods sold}}{\text{Inventory}}$	3.24	2.89	—	—
(15) Fixed Asset Turnover	$\frac{\text{Sales}}{\text{Fixed assets}}$	1.33	1.00	—	—

(*Continued*)

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Table 2.5. (*Continued*)

Ratio Classification	Formula	JNJ 2011	JNJ 2012	Industry 2008	Industry 2009
(16) Total Asset Turnover	$\frac{\text{Sales}}{\text{Total assets}}$	0.57	0.55	—	—
(17) Networking-Capital Turnover	$\frac{\text{Sales}}{\text{Net working-capital}}$	2.06	3.08	—	—
Profitability Ratios					
(18) Profit Margin	$\frac{\text{Net income}}{\text{Sales}}$	14.87%	16.14%	—	—
(19) ROA	$\frac{\text{Net income}}{\text{Total assets}}$	8.51%	8.94%	—	—
(20) ROE	$\frac{\text{Net Income}}{\text{Total equity}}$	16.94%	16.74%	—	—
Market Value Ratios					
(21) Price-earnings ratio	$\frac{\text{Market price per share}}{\text{Earnings per share}}$	18.53	17.63	—	—
(22) Market-to-book ratio	$\frac{\text{Market price per share}}{\text{Book value per share}}$	1.58	1.58	—	—
(23) Earnings yield	$\frac{\text{Earnings per share}}{\text{Market price per share}}$	5.40%	5.67%	—	—
(24) Dividend yield	$\frac{\text{Dividend per share}}{\text{Market price per share}}$	3.43%	3.45%	—	—
(25) PEG ratio**	$\frac{\text{Price-earnings ratio}}{\text{Earnings growth rate (\%)}}$	300.02	269.49	—	—
(26) Enterprise value-Earnings before interest, tax, depreciation and amortization (EBITDA) ratio***	$\frac{\text{Enterprise value}}{\text{EBITDA}}$	13.14	12.96	—	—

*MS is the marketable securities.
**For price-earnings-growth (PEG) ratio, the earnings growth rate is estimated by sustainable growth gate.
***Enterprise value = Total market value of equity + Book value of total liabilities — Cash, EBITDA is earnings before interest, taxes, depreciation, and amortization.
Total market value of equity = market price per share * shares outstanding (basic).
For the earnings per share, we used the basic net earnings per share in JNJ.
Since earnings statement only provide EBT, our Earnings before interest and tax (EBIT) = EBT + interest expense.
Book value per share = total asset/share outstanding (basic).
Price-earnings ratio = market price per share/ earnings per share (basic).

useful to creditors when deciding how much debt they're willing to extend to the asking party. (Investopedia.com)

The networking-capital to total asset ratio (ratio 4) is the NWC divided by the total assets of the company. A relatively low value might indicate relatively low levels of liquidity.

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Long-term Solvency or Financial Leverage Ratios

If an analyst wishes to measure the extent of a firm's debt financing, a leverage ratio (LR) is the appropriate tool to use. This group of ratios reflects the financial risk posture of the firm. The two sources of data from which these ratios can be calculated are the balance sheet and the statement of earnings.

The balance-sheet LR measures the proportion of debt incorporated into the capital structure. The debt–equity ratio measures the proportion of debt that is matched by equity; thus this ratio reflects the composition of the capital structure. The debt–asset ratio (ratio 5), on the other hand, measures the proportion of debt-financed assets currently being used by the firm. Other commonly used LR include the equity multiplier ratio (7) and the time interest paid ratio (8).

Debt-to-equity (6) is a variation on the total debt ratio. It's total debt divided by total equity.

Long-term debt ratio (9) is long-term debt divided by the sum of long-term debt and total equity.

Cash coverage ratio (10) is defined as the sum of EBIT and depreciation divided by interest. The numerator is often abbreviated as EBITDA.

The income-statement LR measures the firm's ability to meet fixed obligations of one form or another. The time interest paid, which is earnings before interest and taxes over interest expense, measures the firm's ability to service the interest expense on its outstanding debt. A more broadly defined ratio of this type is the fixed-charge coverage ratio, which includes not only the interest expense but also all other expenses that the firm is obligated by contract to pay. (This ratio is not included in Table 2.5 because there is not enough information on fixed charges for these firms to calculate this ratio.)

Asset Management or Turnover (Activity) Ratios

This group of ratios measures how efficiently the firm is utilizing its assets. With activity ratios, one must be particularly careful about the interpretation of extreme results in either direction; very high values may indicate

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possible problems in the long term, and very low values may indicate a current problem of low sales or not taking a loss for obsolete assets. The reason that high activity may not be good in the long term is that the firm may not be able to adjust to an even higher level of activity and therefore may miss out on a market opportunity. Better analysis and planning can help a firm get around this problem.

The days-in-accounts-receivable or average collection-period ratio (11) indicates the firm's effectiveness in collecting its credit sales. The other activity ratios measure the firm's efficiency in generating sales with its current level of assets, appropriately termed turnover ratios. While there are many turnover ratios that can be calculated, there are three basic ones: inventory-turnover (14), fixed assets turnover (15), and total assets turnover (16). Each of these ratios measures a quite different aspect of the firm's efficiency in managing its assets.

Receivables turnover (12) is computed as credit sales divided by accounts receivable. In general, a higher accounts receivable turnover suggests more frequent payment of receivables by customers.

In general, analysts look for higher receivables turnover and shorter collection periods, *but this combination may imply that the firm's credit policy is too strict*, allowing only the lowest risk customers to buy on credit. Although this strategy could minimize credit losses, it may hurt overall sales, profits, and shareholder wealth.

Day's sales in inventory ratio (13) estimates how many days, on average, a product sits in the inventory before it is sold.

Net working-capital turnover (17) measures how much "work" we get out of our working-capital.

Profitability Ratios

This group of ratios indicates the profitability of the firm's operations. It is important to note here that these measures are based on past performance. Profitability ratios are generally the most volatile, because many of the variables affecting them are beyond the firm's control. There are three groups of profitability ratios; those measuring margins, those measuring returns, and those measuring the relationship of market values to book or accounting values.

Profit-margin ratios show the percentage of sales dollars that the firm was able to convert into profit. There are many such ratios that can be calculated to yield insightful results, namely: profit margin (18), ROA (19), and return on equity (ROE) (20).

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Return ratios are generally calculated as a return on assets (ROA) or equity. The ROA ratio (19) measures the profitability of the firm's asset utilization. The ROE ratio (20) indicates the rate-of-return earned on the book value of owner's equity. Market-value analyses include (i) market-value/book-value ratio and (ii) price per share/earnings per share (P/E) ratio. These ratios and their applications will be discussed in Chapter 6.

Overall, all four different types of ratios (as indicated in Table 2.5) have different characteristics stemming from the firm itself and the industry as a whole. For example, the collection-period ratio (which is Accounts Receivable times 365 over Net Sales) is clearly the function of the billings, payment, and collection policies of the pharmaceutical industry. In addition, the fixed-asset turnover ratios for those firms are different, which might imply that different firms have different capacity utilization.

Market Value Ratios

A firm's profitability, risk, quality of management, and many other factors are reflected in its stock and security prices. Hence, market value ratios indicate the market's assessment of the value of the firm's securities.

The PE ratio (21) is simply the market price of the firm's common stock divided by its annual earnings per share. Sometimes called the *earnings multiple*, the PE ratio shows how much the investors are willing to pay for each dollar of the firm's earnings per share. Earnings per share comes from the income statement. Therefore, earnings per share is sensitive to the many factors that affect the construction of an income statement, such as the choice of GAAP to management decisions regarding the use of debt to finance assets. Although earnings per share cannot reflect the value of patents or assets, the quality of the firm's management, or its risk, and stock prices can reflect all of these factors. Comparing a firm's PE ratio to that of the stock market as a whole, or with the firm's competitors, indicates the market's perception of the true value of the company.

Market-to-book ratio (22) measures the market's valuation relative to balance sheet equity. The book value of equity is simply the difference between the book values of assets and liabilities appearing on the balance sheet. The price-to-book-value ratio is the market price per share divided by the book value of equity per share. A higher ratio suggests that investors are more optimistic about the market value of a firm's assets, its intangible assets, and the ability of its managers.

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Earnings yield (23) is defined as earnings per share divided by market price per share and is used to measure return on investment. Dividend yield (24) is defined as dividend per share divided by market price per share, which is used to determine whether this company's stock is an income stock or a gross stock. A gross stock dividend yield is very small or even zero. For example, the stock from a utility industry dividend yield is very high. In addition, in the early stage of Microsoft and Apple computer they didn't even pay cash dividend, therefore their dividend yield was zero.

PEG ratio is defined as price-earnings ratio divided by earnings growth rate. The PEG ratio is used to determine a stock's value while taking the company's earnings growth into account, and is considered to provide a more complete picture than the PE ratio. While a high PE ratio may make a stock look like a good buy, factoring in the company's growth rate to get the stock's PEG ratio can tell a different story. The lower the PEG ratio, the more the stock may be undervalued given its earnings performance. The PEG ratio that indicates an over or underpriced stock varies by industry and by company type, though a broad rule of thumb is that a PEG ratio below one is desirable. Also, the accuracy of the PEG ratio depends on the inputs used. Using historical growth rates, for example, may provide an inaccurate PEG ratio if future growth rates are expected to deviate from historical growth rates. To distinguish between calculation methods using future growth and historical growth, the terms "forward PEG" and "trailing PEG" are sometimes used.

Enterprise value is an estimate of the market value of the company's operating assets, which means all the assets of the firm except cash. Since market values are usually unavailable, we use the right-hand side of the balance sheet and calculate the enterprise value as:

$$\text{Enterprise value} = \text{Total market value} + \text{Book value of all liabilities} - \text{Cash}.$$

Notice that the sum of the value of the market values of the stock and all liabilities equals the value of the firm's assets from the balance sheet identity. Enterprise value is often used to calculate the Enterprise value-EBITDA ratio (26):

$$\text{EBITDA ratio} = \frac{\text{Enterprise value}}{\text{EBITDA}}.$$

This ratio is similar to the PE ratio, but it relates the value of all the operating assets to a measure of the operating cash flow generated by those assets.

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Estimation of the Target of a Ratio

An issue that must be addressed at this point is the determination of an appropriate proxy for the target of a ratio. For an analyst, this can be an insurmountable problem if the firm is extremely diversified, and if it does not have one or two major product lines in industries where industry averages are available. One possible solution is to determine the relative industry share of each division or major product line, then apply these percentages to the related industry averages. Lastly, derive one target ratio for the firm as a whole with which its ratio can be compared. One must be very careful in any such analysis, because the proxy may be extremely over- or underestimated. The analyst can also use Standard Industrial Classification (SIC) codes to properly define the industry of diversified firms. The analyst can then use 3- or 4-digit codes and compute their own weighted industry average.

Often an industry average is used as a proxy for the target ratio. This can lead to another problem, the inappropriate calculation of an industry average, even though the industry and companies are fairly well-defined. The issue here is the appropriate weighing scheme for combining the individual company ratios in order to arrive at one industry average. Individual ratios can be weighed according to equal weights, asset weights, or sales weights. The analyst must determine the extent to which firm size, as measured by asset base or market share, affects the relative level of a firm's ratios and the tendency for other firms in the industry to adjust toward the target level of this ratio. One way this can be done is to calculate the coefficients of variation for a number of ratios under each of the weighing schemes and to compare them to see which scheme consistently has the lowest coefficient variation. This would appear to be the most appropriate weighing scheme. Of course, one could also use a different weighing scheme for each ratio, but this would be very tedious if many ratios were to be analyzed. Note, that the median rather than the average or mean can be used to avoid needless complications with respect to extreme values that might distort the computation of averages. In the dynamic analysis that follows, the equal-weighted average is used throughout.

2.4.2. *Dynamic Analysis of Financial Ratios*

In basic finance and accounting courses, industry norms are generally used to determine whether the magnitude of a firm's financial ratios is acceptable. Taken separately, ratios are mere numbers. This can lead to

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some problems in making comparisons among and drawing conclusions from them. In addition, by making only static, one-ratio-to-another comparisons, we are not taking advantage of all the information they can provide. A more dynamic analysis can improve our ability to compare companies with one another and to forecast future ratios. Regressing CR against past ratios helps one analyze the dynamic nature and the adjustment process of a firm's financial ratio.

Single-Equation Dynamic Adjustment Process

1. Basic Model. Lev (1969) uses the concept of the partial-adjustment model to define a dynamic financial-ratio adjustment process as

$$Y_{j,t} = Y_{j,t-1} + \delta_j(Y^*_{j,t} - Y_{j,t-1}), \tag{2.1}$$

where $0 \leq \delta_j \leq 1$ and

$\delta_j = A$ partial adjustment coefficient;
$Y_{j,t} =$ Firm's jth financial ratio period t;
$Y_{j,t-1} =$ Firm's jth financial ratio period $t-1$; and
$Y^*_{j,i} =$ Firm's jth financial ratio target in period t.

This model is used in a wide variety of empirical applications of the dynamic properties of financial analysis and forecasting, such as investment, financing, and dividend decisions. The relationship postulates that at any time, t, only a fixed fraction of the desired adjustment is achieved in that period. Thus, the coefficient of adjustment, δ_j, reflects the fact that there are limitations to the periodic adjustment of ratios. Therefore, the coefficient of adjustment should be interpreted cautiously.

Lev (1969) suggests that differences across ratios in their speed of adjustment coefficient, δ_j, are a function of two conflicting types of costs: (1) the cost of adjustment and (2) the cost of being out of equilibrium. These two costs must be balanced for each ratio. Equation (2.1) implies that a firm's current financial ratio is equal to the last period's financial ratio plus an adjustment term. The adjustment factor depends upon two elements: the partial adjustment coefficient, δ_i, and the difference between $Y^*_{j,t}$ and $Y_{j,t-1}$. However, $Y^*_{j,t}$ is not an observable variable, so we must find some alternative proxy.

To resolve the problem associated with determining the target ratio ($Y^*_{j,t}$), Lev assumes that: (1) $Y^*_{j,t}$ is exactly equal to the industry average of the jth financial ratio in the previous period, denoted as $X_{j,t-1}$ and

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(2) $Y^*_{j,t}$ is proportional to $X_{j,\,t-1}$, that is $CX_{j,\,t-1}$ where C is the related proportional constant. A generalized proxy of $Y^*_{j,t}$ can be defined as

$$Y^*_{j,t} = CX_{j,\,t-1} + \tau_{j,t}, \tag{2.2}$$

where $0 \leq C \leq 1$ and $\tau_{j,t}$ represents the proxy error. Proxy error is the error arising from the fact that the substitute, or proxy, ratio only partially approximates the desired target ratio. If $C = 1$ and $\tau_{j,t} = 0$, then $X_{j,\,t-1}$ is the perfect proxy for $Y^*_{j,t}$. We can then substitute $X_{j,\,t-1}$ for $Y^*_{j,t}$ in Eq. (2.1) and obtain:

$$Y_{j,t} - Y_{j,\,t-1} = \delta_j[X_{j,\,t-1} - Y_{j,\,t-1}]. \tag{2.3}$$

In order to estimate the partial adjustment coefficient, δ_j, a simple time-series regression can be run and used in the empirical study. The linear form of this regression is defined as

$$Z_{j,t} = A_j + B_j W_{j,\,t-1} + \varepsilon_{j,t}, \tag{2.4}$$

where

$$Z_{j,t} = Y_{j,t} - Y_{j,\,t-1};$$

$$W_{j,\,t-1} = X_{j,\,t-1} - Y_{j,\,t-1};$$

$$A_j \text{ and } B_j = \text{regression parameters}$$

and

$$\varepsilon_{j,t} = \text{the error term.}$$

2. Extensions of this model. Lev also suggests a log-linear form of this model in order to study the dynamic ratio-adjustment process:

$$Z\prime_{j,t} = A\prime_j + B\prime_j W\prime_{j,\,t-1} + \varepsilon\prime_{j,t}, \tag{2.5}$$

where

$$Z'_{j,t} = \log(Y_{j,t}) - \log(Y_{j,\,t-1}),$$

$$W'_{j,\,t-1} = \log(X_{j,\,t-1}) - \log(Y_{j,i-1})$$

and

$$\varepsilon'_{j,t} = \text{the error term.}$$

One of the possible advantages of the log-linear form of this model over the linear form is that the estimated B'_j represents the elasticity of change,

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while the estimated B_j does not. The argument is based upon the fact that

$$B'_j = \frac{\partial \log\left(\dfrac{Y_{j,t}}{Y_{j,\,t-1}}\right)}{\partial \log\left(\dfrac{X_{j,\,t-1}}{Y_{j,\,t-1}}\right)} = \frac{\%\text{ change in }\left[\dfrac{\mathrm{Y_{j,t}}}{Y_{j,\,t-1}}\right]}{\%\text{ change in }\left[\dfrac{\mathrm{X_{j,\,t-1}}}{Y_{j,\,t-1}}\right]}. \tag{2.6}$$

This model can be generalized by assuming that the optimal ratio level attained by the firm is last period's industry ratio average times an adjustment factor, as follows:

$$Y^*_{j,t} = CX_{j,\,t-1}. \tag{2.7}$$

The adjustment coefficient, C, indicates that firms tend to maintain a fixed deviation from the industry mean in their adjustment process. Furthermore, the analysis of the coefficient and the partial adjustment coefficient (δ_j) should be very helpful in demonstrating the dynamic nature of a firm's financial structure, its financial ratios, and their adjustment toward the industry mean.

As for predicting future ratios, Lev finds that the model's predictive powers can be enhanced substantially through the following extension to multiple regression:

$$Y_{j,t} = \hat{A} + \hat{B}_1 X_{j,\,t-1} + \hat{B}_2 Y_{j,\,t-1} + \varepsilon_{j,t}. \tag{2.8}$$

This model is found to be substantially more accurate in the prediction of future ratios, while the model detailed in Eq. (2.4) is better at estimating the partial adjustment coefficient, B_j. With this model, analysts can forecast future possibilities. Furthermore, once the future ratios are estimated, one can work backward and determine the estimated levels of individual accounts, which facilitates planning ahead to meet unpleasant future economic situations.

3. Empirical Data. Annual financial ratio data for JNJ and the industry as a whole will be used to show how Eqs. (2.4) and (2.5) perform in empirical financial ratio analysis. The sample period from 1992 to 2004 for JNJ in terms of Eq. (2.5) has been empirically estimated. The estimated B'_j and A'_j and the information needed to estimate these two regression parameters are

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Table 2.6. Dynamic Adjustment Ratio Regression Results.

Variable	CR	LR
Mean Z	0.0075	−0.03083
Mean W	−0.14583	0.361666667
Var(Z)	0.013039	0.006099
Cov(Z,W)	0.074	0.009
B_j`	0.810*	0.259
t-statistics	[3.53]	[1.06]
A_j`	0.032	−0.042

*Partial adjustment coefficient significant at 95% level.

Table 2.7. Ratio Correlation Coefficient Matrix.

	CR	AT	GPM	LR
CR	1.0			
AT	−0.443841	1.0		
GPM	0.363273	0.381393	1.0	
LR	−0.51175	0.21961	−0.05028	1.0

listed in Table 2.6. Based upon the procedure discussed in Appendix 2.A, the estimated mean Z, mean W, Var (W), and COV (Z, W) can be used to estimate B'_j and A'_j Estimates of Table 2.6 indicate the partial adjustment coefficients associated with both CR and LR for JNJ.

The accounting general ledger items used in calculating ratios are interrelated. A ratio is merely calculated by using at least two such items. Hence, important financial ratios of a firm may be interrelated. The correlation coefficient matrix among CR, asset turnover (AT), gross profit margin (GPM), and LR for JNJ is presented in Table 2.7 to show the interrelationship. Fisher's z-statistics imply that the sample correlation coefficients between CR and GPM, CR and LR, AT and LR, and GPM and LR are statistically significant at the 95% level. Based upon approximations of Fisher's z-statistics, the sample correlation coefficients (as indicated in Table 2.7) can be statistically tested.[5]

As shown in Table 2.7, the LR is fairly negatively correlated with CR (−0.51175) and GPM (−0.05028) but positively correlated with the activity ratio (AT, 0.21961). Also, GPM is somewhat positively correlated with the CR (0.363273). Thus, it can show that these ratios are interrelated. This tells us that we must use a system of ratios in simple analysis, or use simultaneous equations in a statistical analysis.

[5] $z-$ statistics $= 1/2 \log[(1+\rho)/(1-\rho)]$, where ρ is the simple correlation coefficient.

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2.4.3. *Statistical Distribution of Financial Ratios*

Normal and log-normal distributions are two of the most popular statistical distributions used in accounting and financial analysis. The density function of a normal distribution is

$$F[X] = \frac{1}{\sigma\sqrt{2\pi}} e^{-(X-\mu)^2/2\sigma^2} (-\infty < \mathrm{X} < +\infty), \tag{2.10}$$

where μ and σ^2 are the population mean and variance, respectively, and e and π are given constants; that is, $\pi = 3.14159$ and $e = 2.71828$.

Normal distributions with different means and variances are graphed in Figs. 2.1 and 2.2. In these two figures, $F(X)$ represents the frequency of the variable X. The variance of Fig. 2.1 is larger than the variance of Fig. 2.2.

If a variable is normally distributed, the information that fully describes the distribution is the mean and the variance. The relative skewness of normal distribution is 0, and the kurtosis of a normal distribution is equal to 3.

There is a direct relationship between the normal distribution and the log-normal distribution. If Y is log-normally distributed, then $X = \log Y$ is normally distributed. Following this definition, the mean and the variance of Y can be defined as

$$\mu_Y = \exp\left(\mu_x + \frac{1}{2}\sigma_x^2\right), \tag{2.11a}$$

$$\sigma_Y^2 = \exp(2\mu_x + \sigma_x^2)(\exp(\sigma_x^2) - 1), \tag{2.11b}$$

where exp represents an exponential with base e.

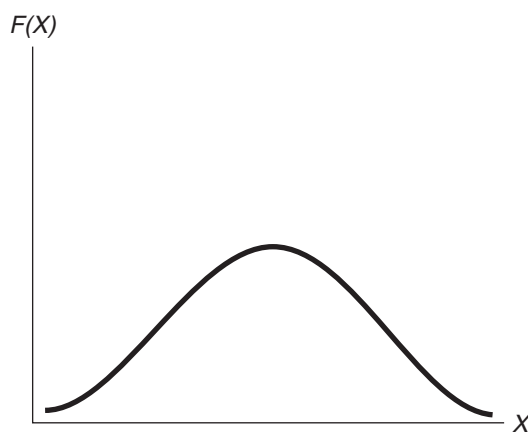


Fig. 2.1. Normal distribution with large variance.

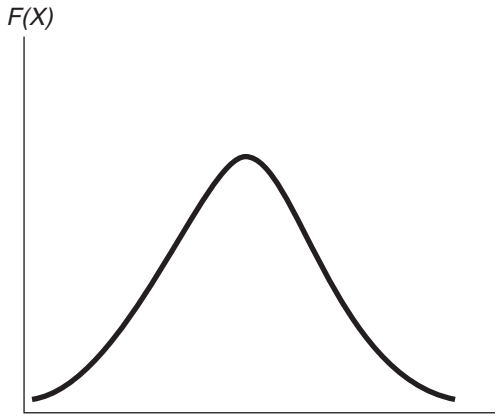


Fig. 2.2. Normal distribution with small variance.

Deakin (1976) finds that the cross-sectional distribution of financial ratios is log-normally distributed instead of normally distributed. Upon analyzing the raw data, he finds that within the pharmaceutical industry, only the debt–asset ratio is normally distributed; this occurs in 15 out of the 19 years of his dataset. All other ratios are log-normally distributed. However, after taking the log transformation, the CR becomes normally distributed in 12 out of the 19 years; while the normality of the debt-asset ratio drops from 15 to only 6 of the 19 years.

Generally, it seems that financial ratios are not normally distributed and that log transformation does help normalize the data in some, but not all, cases. The reason we need a normal distribution for analyzing ratios is in testing the significance of a difference, such as between the behavior of firm and industry figures for the same ratio. It is necessary for the analyst to look at data with and without transformations, in order to determine which set of data fits the normal distribution more closely.

2.5. CVP Analysis and its Applications

CVP analysis is a synthesized analysis of the statement of earnings. Volume, price per unit, variable cost per unit, and the total fixed cost are the key variables for doing this kind of analysis. The basic type of CVP analysis is the break-even analysis, which can be extended to operating and financial leverage analysis. All of these analyses are important tools of financial analysis and control. Technically, ratio-variable inputs are required for

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performing these analyses. Conceptually, CVP and its derived relationships are designed to analyze the statement of earnings in terms of an aggregated ratio indicator. Hence, CVP analysis can be regarded as one kind of financial ratio analysis.

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2.5.1. *Deterministic Analysis*

Deterministic break-even analysis is an important concept in basic microeconomics, accounting, finance, and marketing courses. Mathematically, the operating profit (EBIT) can be defined as

$$\text{Operating Profit} = \text{EBIT} = Q(P - V) - F, \tag{2.12}$$

where

$Q =$ Quantity of goods sold;
$P =$ Price per unit sold;
$V =$ Variable cost per unit sold;
$F =$ Total amount of fixed costs; and
$P - V =$ Contribution margin.

If operating profit is equal to zero, Eq. (2.12) implies that $Q(P-V)-F = 0$ or that $Q(P - V) = F$, that is

$$Q^* = \frac{F}{(P - V)}. \tag{2.13}$$

Equation (2.13) represents the break-even quantity, or that quantity of sales at which fixed costs are just covered. There are two kinds of break-even analysis, linear and non-linear. The two forms are shown graphically in Figs. 2.3 and 2.4.

There are very important economic interpretations of these alternative break-even analyses. Linear representation of the total revenue curve implies that the firm operates within a perfect output or product market. The linear total cost curve implies that the input market is linear or perfect and the return (economies) to scale is constant. If these conditions do not hold, linear break-even analysis becomes either unrealistic or only an approximation of the real situation facing the firm.

In the real world, returns (economies) to scale can either be constant, increasing, or decreasing. A non-linear representation of the variable cost and total revenue curves is a more accurate representation of the real break-even level of sales using this form of analysis.

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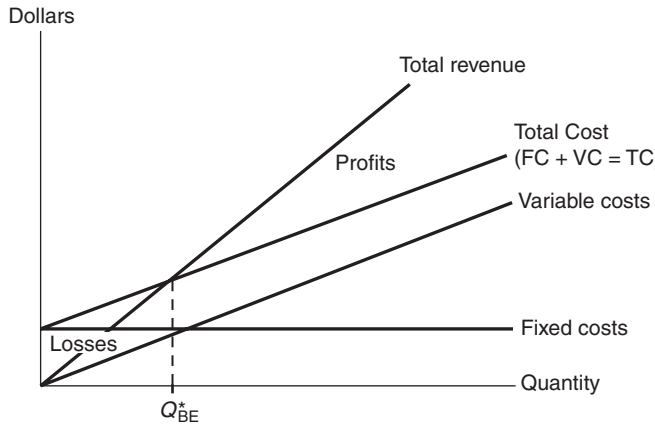


Fig. 2.3. Linear break-even analysis.

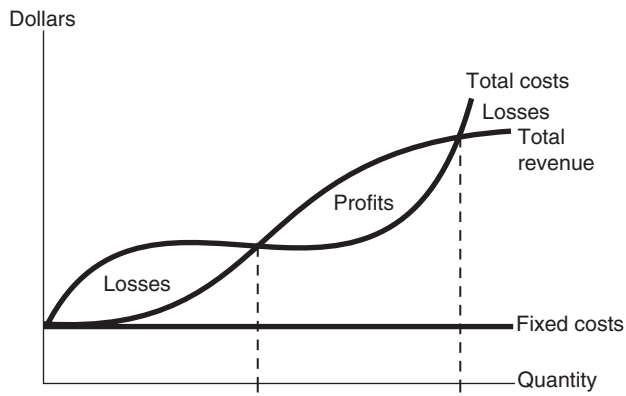


Fig. 2.4. Non-linear break-even analysis.

Break-even analysis can be used in three separate but related ways in financial management. That is: (i) to analyze a program of modernization and automation, (ii) to study the effects of a general expansion in the level of operations, and (iii) in new-product decision. These operating leverage decisions can be defined more precisely in terms of the way a given change in volume affects profits. For this purpose, we use the definition of degree of operating leverage (DOL) represented as Eq. (2.14):

$$\text{DOL} = \frac{\%\ \text{Change in profits}}{\%\ \text{Change in sales}} = \frac{Q(P-V)}{Q(P-V)-F} = 1 + \frac{\text{Fixed Costs}}{\text{Profits}}. \tag{2.14}$$

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The first equality in Eq. (2.14) is the basic definition of DOL. The second equality is obtained by substituting the definition of profits and sales from Eq. (2.12) (see Appendix 6.B for derivation). The third equality implies that the DOL increases with a firm's exposure to fixed costs (see Appendix 6.B for derivation).

Based upon the definition of linear break-even quantity, defined in Eq. (2.13), the DOL can be rewritten as

$$DOL = \frac{1}{[1 - (Q^*/Q)]}. \tag{2.15}$$

There are two important implications of this formulation: (1) if $Q > Q^*$, then DOL > 0, and the change in profits is in the same direction as the change in sales; and (2) if $Q < Q^*$, then DOL < 0 and the change in losses is in the opposite direction from the change in sales (i.e., if sales increase, losses will decrease). Both Eqs. (2.14) and (2.15) can be used to calculate the DOL at any level of output, Q. If company XYZ's break-even quantity Q^* is 50,000 units, then DOL at 100,000 production units is 2. This implies that 1% change of XYZ's sales will generate a 2% change in its profit.

2.5.2. *Stochastic Analysis*

In reality, net profit is a random variable because the quantity used in the analysis should be the quantity sold, which is unknown and random, rather than the quantity produced which is internally determined. This is the simplest form of stochastic CVP analysis; for there is only one stochastic variable and no need to be concerned about independence among variables. The distribution of sales is shown graphically in Fig. 2.5.

A slightly more complicated form of stochastic CVP analysis is obtained when it is assumed that both the quantity of good sold (Q) and the contribution margin ($P - V$) are stochastic variables and are independently distributed. The independence assumption is reasonable because the second stochastic variable is defined as the contribution margin, rather than the three separate random variables Q, P, and V. In this situation, quantity and price are probably not independent, because both distributions are determined by imperfections in the product market. Furthermore, these three variables are generally highly negatively correlated, while the normality of their distributions is questionable (needs further empirical testing). Under the contribution margin approach, one variable that is subtracted from

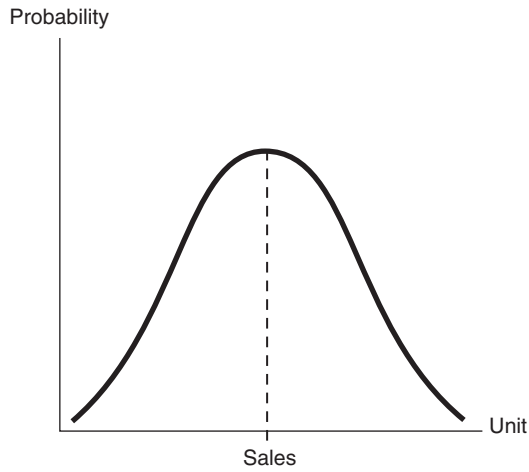


Fig. 2.5. Distribution of sales.

prices, variable costs, has a distribution that is determined by imperfections in the input market. This drastically reduces the degree of correlation with the quantity sold.

While the assumption of independence seems to be fairly accurate in some cases, there is need for more research in this area. Hilliard and Leitch (1975) addressed this problem through the use of a log-normal approach to CVP analysis under uncertainty. They have developed an easily implemented model that can handle dependence among input variables, and that should prove to be helpful to the analyst.

Equation (2.12) can be generalized to allow Q, P, and V to be stochastic variables. Profits also become a stochastic variable, as represented in Eq. (2.16):

$$\text{Operating profit} = \text{EBIT} = Q^{\pi}(P^{\pi} - V^{\pi}) - F. \tag{2.16}$$

This formulation for CVP analysis under uncertainty is very important for the decision-maker. It is becoming increasingly important to move from point (certainty) to interval (uncertainty) estimates, particularly in today's dynamic business environment, the stochastic nature of this CVP formulation allows one to calculate the point estimate and also the related interval estimate based on the combined probability distributions. This application of probabilistic models greatly enhances the degree of realism. Uncertainty break-even analysis has been generalized by Wei (1979) and Yunker (1982)

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to take into account the crucial elements of random demand and the level of production in the determination of actual sales and resulting profits.

Besides the applications discussed in this section, both CVP and break-even analysis can be integrated with the net present value method of capital budgeting decisions to do financial analysis. Reinhardt (1973) uses this approach to do break-even analysis for Lockheed's Tri Star Program. The Lockheed Tri Star Program uses net present value type of break-even analysis and will be explained later in detail in Chapter 13. The major difference between the net present value (NPV) type of break-even analysis and the "naive" break-even analysis is that the latter does not take into account the cost of capital. This will be explored in the capital budgeting under certainty chapter in detail.

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2.6. Accounting Income Versus Economic Income[6]

There are three ways of describing income: accounting earnings, economic income, and finance income. Accounting earnings are based upon recording transactions according to GAAP using accruals and deferrals, rather than cash flows. Accountants measure the change in stockholders' wealth due to the operations of the firm. Economic earnings are more abstract than accounting earnings. Economic income measures the change in wealth based upon both realized and unrealized gains and losses.

Finance income is based on cash flow changes in wealth. Cash income is cash revenues less cash expenses. Finance income defines changes in wealth as net cash flow, or net cash income less cash investment outlays. Technically, economic income, rather than accounting earnings, should be used to determine the value of a firm. However, theoretical economic income is not directly observable, so accounting earnings are generally used as proxy for them. Conceptually, the relationship between accounting earnings and economic income can be defined as

$$E_t = A_t + P_t, \tag{2.17}$$

where

E_t = Economic income,
A_t = Accounting earnings, and
P_t = Proxy errors.

[6]This section is based, in part, on Haley and Schall (1979, pp. 8–10).

In estimating the cost of capital for the utility industry, Miller and Modigliani (1966) take accounting earnings as a proxy for economic income, using instrumental-variable technique, to remove the proxy errors. Instrumental-variable technique can be found in Appendix 25.A of this book. Finance income (cash flows instead of accounting income) will be used to evaluate alternative capital budgeting decisions, which are discussed in Chapters 9 and 10. The main difference between accounting income and cash flows (finance income) is that the former does not focus on cash flows when they occur, while the latter does. The cash flows, for example, correctly deduct the entire expenditure for investment in plant and equipment at the time the cash outflows occur.

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2.7. Summary

In this chapter, the usefulness of accounting information in financial analysis is conceptually and analytically evaluated. Both statistical methods and regression analysis techniques show how accounting information can be used to perform active financial analysis for the pharmaceutical industry.

In these analyses, static ratio analysis is generalized to dynamic ratio analysis. In addition, both deterministic and stochastic CVP analyses are examined. The potential applications of CVP analysis in financial analysis and planning are discussed in some detail. Overall, this chapter gives readers a good understanding of basic accounting information and econometric methods, which are needed for financial analysis and planning.

Problem Set

1. What is the value of accounting information in the process of financial analysis and planning? What are the components to annual and quarterly accounting information and their respective advantages and disadvantages for financial analysis and planning?
2. Define the following terms:
   (a) Real versus financial market
   (b) M_1 and M_2
   (c) Leading economic indicators
   (d) NYSE, AMEX, and OTC
   (e) Primary versus secondary stock market
   (f) Bond market
   (g) Options and futures markets.

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3. What are limits related to the use of accounting information? What are differences that can result from using accounting, economics, and finance information? How can these limits and differences be overcome?
4. Discuss the major difference between linear and non-linear break-even analysis.
5. What can ratio analysis contribute to financial planning and analysis? What are the determinants of the dynamic adjustment process? How are target ratios derived?
6. What is the relationship between CVP and ratio analysis? How can it be used in financial analysis and planning?
7. Briefly discuss the issue related to net income (EAIT) in financial analysis and planning.
8. A market model is defined as

$$R_{jt} = A_j + B_j R_{mt} + \in_j,$$

where R_{jt}, R_{mt} are the rates-of-return for jth security and market rates-of-return respectively; A_j is the intercept and B_j is the slope. Use the Ordinary Least Squares (OLS) theory and method discussed in Appendix A of this chapter to show how A_j and B_j can be estimated.

9. Parts of financial records for Company XYZ are:

Anticipated sales = \$400,000
Degree of financial leverage = 4/3
Variable cost = \$200,000
Combined leverage effect = 2
Quantity sold = \$100,000 units
Profit margin = 5%
Total debt = \$200,000
Leverage ratio = 1/2
Common stock outstanding = 10,000 shares
Current price per share = \$40
Retention rate = 1/2
IRR = 8%

Calculate the following:

(a) The total interest expense.
(b) The total fixed cost.
(c) The degree of operating leverage.
(d) The break-even quantity.

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(e) The EAIT.
(f) Total corporation income tax expense.
(g) The return on net worth.
(h) The return on total asset.
(i) The EPS.
(j) The P/E ratio for common stock.
(k) The pay-out ratio for dividend.
(l) The growth rate for the common stock.
(m) The required rate-of-return.
(n) Total asset turnover.
(o) Analyze the financial situation of Company XYZ.
(p) If the probabilistic concepts are applied to break-even analysis, how should the analyses in (o) be revised?

10. Use the ratio information listed in Table 2.5 of the text to estimate the partial ratio adjustment model for GM and interpret the related results.
11. ABC Company's financial records are as follows:

Quantity of goods sold = 10,000
Price per unit sold = $20
Variable cost per unit sold = $10
Total amount of fixed cost = $50,000
Corporate tax rate = 50%

(a) Calculate EAIT.
(b) What is the break-even quantity?
(c) What is the DOL?
(d) Should the ABC Company produce more for greater profits?

12. ABC Company's predictions for next year are as follows:

	Probability	**Quantity**	**Price ($)**	**Variable Cost/Unit ($)**	**Corporate Tax Rate**
State 1	0.3	1,000	10	5	0.5
State 2	0.4	2,000	20	10	0.5
State 3	0.3	3,000	30	15	0.5

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In addition, we also know that the fixed cost is \$15,000. What is next year's expected EAIT?

13. Use an example to discuss four alternative depreciation methods.
14. XYX, Inc. currently produces one product that sells for \$330 per unit. The company's fixed costs are \$80,000 per year; variable costs are \$210 per unit. A salesman has offered to sell the company a new piece of equipment which will increase fixed costs to \$100,000. The salesman claims that the company's break-even number of units sold will not be altered if the company purchases the equipment and raises its price (assuming variable costs remain the same).
    (a) Find the company's current break-even level of units sold.
    (b) Find the company's new price if the equipment is purchased and prove that the break-even level has not changed.
15. Consider the following financial data of a corporation:

    Sales = \$500,000
    Quantity = 25,000
    Variable cost = \$300,000
    Fixed cost = \$50,000

    (a) Calculate the DOL at the above quantity of output.
    (b) Find the break-even quantity and sales levels.
16. On the basis of the following firm and industry norm ratios, identify the problem that exists for the firm:

Ratio	**Firm**	**Industry**
Total Asset Utilization	2.0	3.5
Average Collection Period (days)	45	46
Inventory-Turnover (times)	6	6
Fixed Asset Utilization	4.5	7.0

17. The financial ratios for Wallace, Inc., a manufacturer of consumer household products, are given below along with the industry norm:

	Firm			
Ratio	**1986**	**1987**	**1988**	**Industry**
Current Ratio	1.44	1.31	1.47	1.2
Quick Ratio	0.66	0.62	0.65	0.63
Debt to Assets	0.56	0.54	0.46	0.50
Debt to Equity	1.24	1.14	0.84	1.00
Times Interest Earned	2.75	5.57	7.08	5.00
Day's Sales in Receivable (Average Collection Period)	33	37	32	34
Inventory-Turnover	7.86	7.62	7.72	7.6
Fixed Asset Turnover	2.60	2.44	2.56	2.80
Total Asset Turnover	1.24	1.18	2.8	1.20
Profit Margin	0.02	0.05	0.05	0.05
ROA	0.02	0.06	0.07	0.06
ROE	0.06	0.12	0.12	0.13

Analyze Merck's ratios over the 3-year period for each of the following categories:

(a) Liquidity
(b) Asset utilization
(c) Financial leverage
(d) Profitability

18. Below are the Balance Sheet and the Statement of Earnings for Johnson and Johnson:

Balance Sheet for Johnson and Johnson on 12/30/12	
Assets	
Cash and Cash Equivalent ($)	$14,911
Marketable Securities	6,178

(*Continued*)

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(*Continued*)

Account Receivable	11,309
Inventory	7,495
Deferred Taxes on Income	3,139
Prepaid Expenses and Other Receivable	3,084
Total Current Assets	$46,116
Marketable Securities — Non-current	—
PP&E, Net	16,097
Intangible Assets, Net	51,176
Deferred Taxes on Income	4,541
Other Assets	3,417
Total Assets	$121,347
Liabilities and Stockholder's Equity	
Loans and Notes Payable	$4,676
Account Payable	5,831
Accrued Liabilities	7,299
Accrued Rebates, Returns, and Promotion	2,969
Accrued Salaries, Wages, and Commissions	2,423
Taxes on Income	1,064
Total Current Liabilities	$24,262
Long-Term Debt	11,489
Deferred Tax Liability	3,136
Employee Related Obligations	9,082
Other Liabilities	8,552
Stockholder's Equity	
Preferred Stock–without Par Value	—
Common Stock — Par Value $1.00	3,120
Net Receivable from Employee Stock Plan	—
Accumulated Other Comprehensive Income	−5,810
Retained Earnings	85,992
Less: Common Stock Held in Treasury	18,476
Total Stockholder's Equity	$64,826
Total Liabilities and Stockholder's Equity	$121,347

(*Continued*)

(*Continued*)

Statement of Earnings for Johnson and Johnson for Year Ending 12/30/12

Sales to Customers	$67,224
Cost of Products Sold	21,658
Gross Profit	45,566
Selling, Marketing, and Administrative expense	20,869
R&D Expense	7,665
In-process R&D	1,163
Interest income	−64
Interest Expense, net of	532
Other (income) expense, Net	1,626
Earnings before Provision for Taxes on Income	$13,775
Provision for Taxes on Income	3,261
Net Earnings	$10,853

(a) Calculate the following ratios for Johnson and Johnson.

		J&J	*Industry*
(1)	Current Ratio	1.9	3.40
(2)	Quick Ratio	1.34	2.43
(3)	Cash Ratio	0.87	
(4)	Net working-capital to total asset	0.18	
(5)	Debt-to-Asset	0.45	88.65
(6)	Debt-to-Equity	0.84	6.46
(7)	Equity Multiplier	1.87	4.41
(8)	Times Interest Paid	26.89	1.12
(9)	Log-term Debt Ratio	0.15	0.34
(10)	Cash Coverage Ratio	33.78	5.25
(11)	Day's Sales in Receivables (Average Collection Period)	61.40	12.00
(12)	Receivable Turnover	5.94	0.12
(13)	Day's Sales in Inventory	126.31	0.18
(14)	Inventory-Turnover	2.89	0.12

(*Continued*)

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(*Continued*)

		J&J	*Industry*
(15)	Fixed Asset Turnover	1.00	
(16)	Total Asset Turnover	0.55	
(17)	Net Working-Capital Turnover	3.08	
(18)	Profit Margin	16.14%	
(19)	ROA	8.94%	
(20)	ROE	16.74%	
(21)	Price-earnings ratio	17.63	
(22)	Market-to-book ratio	1.58	
(23)	Earnings yield	5.67%	
(24)	Dividend yield	3.45%	
(25)	PEG Ratio	269.49	
(26)	Enterprise value — EBITDA ratio	12.96	

(b) Identify Johnson and Johnson's strengths and weaknesses relative to the industry norm.

19. **Please describe the procedure of using the indirect method to compile the cash flow statement based upon the 2011 financial data of JNJ. (Please refer to Appendix 2 of this Chapter to answer this question).**

Appendix 2.A: Simple and Multiple Regression

2.A.1. *Introduction*

Algebra, basic calculus, and linear regression topics are usually required for both undergraduate and graduate business students. This appendix reviews simple regression and multiple regression. This appendix enables the reader to use the available standard multiple regression and simultaneous equation statistic packages (i.e., North Carolina State University's SAS and University of Illinois SOUPAC) to estimate and interpret the empirical results of financial analysis and planning.

2.A.2. *Simple Regression*

If we want to explain the variation of the CR for JNJ Company in period t $[Y_t]$, given the auto industry's CR in period $t-1[X_{t-1}]$, we can choose

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either a linear or a log-linear model, as defined below:

$$Y_t = a + bX_{t-1} + \varepsilon_t, \tag{2A.1a}$$

$$\log Y_t = a' + b' \log X_{t-1} + \varepsilon'_t, \tag{2A.1b}$$

where ε_t and ε'_t are error terms representing the difference between actual CRs and the predicted CRs. The choice between Eqs. (2A.1a) and (2A.1b) depends on whether the related variables, Y_t and X_{t-1}, are normally or log-normally distributed.

In order to obtain the best linear model to predict Y_t, given X_{t-1}, we want to find the equation that minimizes the squared error terms. The error terms ($\hat{\varepsilon}_t$) represent the difference between the actual and current ratio (Y_t) and the current ratio predicted by the estimated linear model ($\hat{Y}_t$). The estimated current ratio can be defined as $Y = \hat{a} + \hat{b}X_{t-1}$. The theory and method of estimating a and b are one of the main issues of this appendix.

In Eq. (2A.1a), if the unconditional probability distribution of Y_t is normally distributed with mean $\overline{Y}$ and variance σ_Y^2, the conditional probability distribution of $Y_t, \tilde{Y}_t | X_{t-1}$ will be a normal distribution with mean $(a + b\overline{X}_{t-1})$ and variance σ_ε^2.

Applying the variance operator to both sides of Eq. (2A.1a) we have

$$\begin{aligned} Var[Y_t] &= Var[a + bX_{t-1} + \varepsilon_t] \\ &= Var[a] + Var[bX_{t-1}] + Var[\varepsilon_t] + 2\text{cov}[a, bX_{t-1}] \\ &\quad + 2\text{cov}[a, \varepsilon_t] + 2\text{cov}[bX_{t-1}, \varepsilon_t], \end{aligned} \tag{2A.2a}$$

$$Var\,[Y_t] = b^2 Var\,[X_{t-1}] + Var\,[\varepsilon_t]\,. \tag{2A.2b}$$

The conditions for obtaining Eq. (2A.2b) are

(i) Both a and b must be constants. Equation (2A.1) is a fixed coefficient model instead of a random coefficient model.
(ii) X_{t-1} must be uncorrelated with ε_t. This implies that there is no errors-in-variables problem inherent in the raw data associated with the independent variable.

Equation (2A.2b) implies that the total variance Y_t, Var$[Y_t]$, can be decomposed into two components: (1) explained variance, $b^2 Var[X_{t-1}]$ and (2) unexplained variance, $Var[\varepsilon_t]]$. If X_{t-1} has any explanatory power, then

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Var$[Y_t]$ will be larger than $Var[\varepsilon_t]$. The degree of explanatory power X_{t-1} can be defined as

$$R^2 = \frac{\text{Variation explained by the explanatory variable}}{\text{Total variation in the dependent variable}} = \frac{b^2 Var\left[X_{t-1}\right]}{Var\left[Y_t\right]}. \tag{2A.3}$$

To obtain the best estimates of a and b, we want to find the equation that minimizes the error sum of squares (ESS) as defined below:

$$ESS = \sum_{t=1}^{n} \left[Y_t - \hat{Y}_t\right]^2 = \sum_{t=1}^{n} \left[Y_t - \hat{a} - \hat{b}X_{t-1}\right]^2. \tag{2A.4}$$

ESS evaluates the predicted relationship by summing the squares of the differences between actual and predicted values of the dependent variable, Y_t. In order to minimize the value of ESS, we must take the derivative of ESS with respect to $\hat{a}$ and $\hat{b}$ and set the results equal to zero, as shown in Eqs. (2A.5a) and (2A.5b):

$$\frac{\partial(ESS)}{\partial a} = -2\sum_{t=1}^{n}\left(Y_t - \hat{a} - \hat{b}X_{t-1}\right) = 0, \tag{2A.5a}$$

$$\frac{\partial(ESS)}{\partial b} = -2\sum_{t=1}^{n} X_{t-1}\left(Y_t - \hat{a} - \hat{b}X_{t-1}\right) = 0. \tag{2A.5b}$$

In order to jointly solve for estimates of the parameters a and b, Eqs. (2A.5a) and (2A.5b) are used to formulate a two-equation simultaneous system as follows:

$$\hat{a}n + \hat{b}\sum_{t=1}^{n} X_{t-1} = \sum_{t=1}^{n} Y_t, \tag{2A.6a}$$

$$\hat{a}\sum_{t=1}^{n} X_{t-1} + \hat{b}\sum_{t=1}^{n} X_{t-1}^2 = \sum_{t=1}^{n} X_{t-1}Y_t. \tag{2A.6b}$$

In these normal equations, two parameters are to be solved for from the given information: $\sum_{t=1}^{n} X_{t-1}; \sum_{t=1}^{n} Y_t; \sum_{t=1}^{n} X_{t-1}^2$ and $\sum_{t=1}^{n} X_{t-1}Y_t$. By

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Cramer's rule (see also Chapter 3), we have

$$\hat{b} = \frac{\begin{vmatrix} n & \sum_{t=1}^{n} Y_t \\ \sum_{t=1}^{n} X_{t-1} & \sum_{t=1}^{n} X_{t-1}Y_t \end{vmatrix}}{\begin{vmatrix} n & \sum_{t=1}^{n} X_{t-1} \\ \sum_{t=1}^{n} X_{t-1} & \sum_{t=1}^{n} X_{t-1}^2 \end{vmatrix}} = \frac{n\left(\sum_{t=1}^{n} X_{t-1}Y_t\right) - \left(\sum_{t=1}^{n} X_{t-1}\sum_{t=1}^{n} Y_t\right)}{n\sum_{t=1}^{n} X_{t-1}^2 - \left(\sum_{t=1}^{n} X_{t-1}\right)^2}, \tag{2A.7a}$$

$$\hat{b} = \frac{Cov[X_{t-1}, Y_t]}{Var[X_{t-1}]}, \tag{2A.7b}$$

$$\begin{aligned} \hat{a} &= \frac{\begin{vmatrix} \sum Y & \sum X_{t-1} \\ \sum X_{t-1}Y_t & \sum X_{t-1}^2 \end{vmatrix}}{\begin{vmatrix} n & \sum X_{t-1} \\ \sum X_{t-1} & \sum X_{t-1}^2 \end{vmatrix}} \\ &= \frac{\left(\sum_{t=1}^{n} Y_t\right)\left(\sum_{t=1}^{n} X_{t-1}^2\right) - \left(\sum_{t=1}^{n} X_{t-1}\right)\left(\sum_{t=1}^{n} X_{t-1}Y_t\right)}{n\sum_{t=1}^{n} X_{t-1}^2 - \left(\sum_{t=1}^{n} X_{t-1}\right)^2} \\ &= \frac{\left(\sum_{t=1}^{n} Y_t/n\right)\left[n\left(\sum_{t=1}^{n} X_{t-1}^2\right) - \left(\sum_{t=1}^{n} X_{t-1}\right)^2\right]}{} \\ &\quad \frac{- \left(\sum_{t=1}^{n} X_{t-1}/n\right)\left[n\left(\sum_{t=1}^{n} X_{t-1}Y_t\right) - \left(\sum_{t=1}^{n} X_{t-1}\right)\left(\sum_{t=1}^{n} X_{t-1}\right)\right]}{n\sum_{t=1}^{n} X_{t-1}^2 - \left(\sum_{t=1}^{n} X_{t-1}\right)^2}, \end{aligned} \tag{2A.8a}$$

$$\hat{a} = \overline{Y} - \overline{X}\hat{b}. \tag{2A.8b}$$

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Equations (2A.7b) and (2A.8b) can be used to estimate the parameters of Eq. (2A.1a). For JNJ's CR and the related industry average data, we have:

$$\begin{aligned} &\overline{X} = 1.730 && Var[X_{t-1}] = 0.0481 \\ &\overline{Y} = 1.587 && Var[Y_{t-1}] = 0.0967 \\ &Cov[Y_t, X_{t-1}] = 0.0402 \\ &b = 0.8358 && \hat{a} = 0.1411 \end{aligned}$$

Before the estimated $\hat{a}$ and $\hat{b}$ are used, they should be tested to determine whether they are statistically different from zero or not. To perform the null hypothesis test, the variance of $\hat{b}$ and $\hat{a}$ should be derived.

Variance of $\hat{b}$

Equation (2A.7b) implies that

$$\hat{b} = \sum_{t=1}^{n} \frac{(x_{t-1} y_t)}{\sum_{t=1}^{n} x_{t-1}^2} = \sum_{t=1}^{n} W_{t-1} y_t, \tag{2A.7c}$$

where

$$\begin{aligned} x_{t-1} &= X_{t-1} - \overline{X} \\ y_t &= Y_t - \overline{Y} \\ W_{t-1} &= \frac{x_{t-1}}{\sum_{t=1}^{n} x_{t-1}^2} \end{aligned}$$

Substituting $y_t = b x_{t-1} + \varepsilon_t$ into Eq. (2A.7b), we have

$$\hat{b} = \sum_{t=1}^{n} W_{t-1} b x_{t-1} + \sum_{t=1}^{n} W_{t-1} \varepsilon_t. \tag{2A.7d}$$

Through the application of the variance operator on Eq. (2A.7d), we have

$$\begin{aligned} Var(\hat{b}) &= E(\hat{b} - b)^2 \\ &= E\left(\sum_{t=1}^{n} W_{t-1} b X_{t-1} + \sum_{t=1}^{n} W_{t-1} \varepsilon_t - b \right)^2 \end{aligned}$$

$$= E\left[\left(\sum_{t=1}^{n} W_{t-1}x_{t-1} - 1\right) b + \sum_{t=1}^{n} W_{t-1}\varepsilon_t\right]^2$$

$$= E\left(\sum_{t=1}^{n} W_{t-1}\varepsilon_t\right)^2, \quad \text{since } \sum_{t=1}^{n} W_{t-1}x_{t-1} = 1.$$

Therefore,

$$Var(\hat{b}) = E[(W_0\varepsilon_1)^2 + 2(W_0W_1\varepsilon_1\varepsilon_2) + (W_1\varepsilon_2)^2 + \cdots]. \tag{2A.9}$$

If ε_t is serially uncorrelated, that is, $E[\varepsilon_t, \varepsilon_{t-1}] = 0$, then Eq. (2A.9) implies that

$$Var(\hat{b}) = E(W_0\varepsilon_1)^2 + E(W_1\varepsilon_2)^2 + \cdots$$
$$= W_0^2E(\varepsilon_1^2) + W_1^2E(\varepsilon_2^2) + \cdots .$$

In generalized form,

$$Var(\hat{b}) = \sum_{t=1}^{n} W_{t-1}^2E(\varepsilon_t^2)$$
$$= \sigma_\varepsilon^2 \sum_{t=1}^{n} W_{t-1}^2$$

But

$$\sum_{t=1}^{n} W_{t-1}^2 = \frac{\sum_{t=1}^{n} x_{t-1}^2}{\left(\sum_{t=1}^{n} x_{t-1}^2\right)^2} = \frac{1}{\sum_{t=1}^{n} x_{t-1}^2}.$$

Therefore,

$$Var(\hat{b}) = \frac{\sigma_\varepsilon^2}{\sum_{t=1}^{n} x_{t-1}^2}. \tag{2A.10}$$

Using a similar derivation, we can derive $Var(\hat{a})$ and $Cov(\hat{a}, \hat{b})$:

$$Var(\hat{a}) = \sigma_\varepsilon^2 \frac{\sum_{t=1}^{n} x_{t-1}^2}{n\sum_{t=1}^{n} x_{t-1}^2}. \tag{2A.11}$$

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$$Cov(\hat{a}, \hat{b}) = -\sigma_{\varepsilon}^2 \frac{\overline{X}}{\sum_{t=1}^{n} x_{t-1}^2}. \tag{2A.12}$$

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Multiple Regression

If the current ratio for JNJ in period $t[Y_t]$ is a function of the pharmaceutical industry's current ratio in period $t-1[X_{1,t-1}]$ and the pharmaceutical industry's LR in period $t-1[X_{2,t-1}]$, the linear relationship can be defined as

$$Y_t = a + bX_{1,\,t-1} + cX_{2,\,t-1} + \varepsilon_t. \tag{2A.13a}$$

The ESS can be defined as

$$ESS = \sum \hat{\varepsilon}_t^2 = \sum (Y_t - \hat{Y}_t)^2,$$

where

$$\hat{Y}_t = \hat{a}_t + \hat{b}X_{1,\,t-1} + \hat{c}X_{2,\,t-1}.$$

To obtain the least-squares estimates of the parameters a, b, and c, we can minimize ESS by calculating its partial derivatives with respect to these three unknown parameters, equating each to zero, and solving:

$$\frac{\partial ESS}{\partial a} = 0 \quad \text{or} \quad \sum Y_t = na + b\sum X_{1,\,t-1} + c\sum X_{2,\,t-1}, \tag{2A.14a}$$

$$\frac{\partial ESS}{\partial b} = 0 \quad \text{or} \quad \sum X_{1,\,t-1}Y_t = a\sum X_{1,\,t-1} + b\sum X_{1,\,t-1}^2 + c\sum X_{1,\,t-1}X_{2,\,t-1}, \tag{2A.14b}$$

$$\frac{\partial ESS}{\partial c} = 0 \quad \text{or} \quad \sum X_{2,\,t-1}Y_t = a\sum X_{2,\,t-1} + b\sum X_{1,\,t-1}X_{2,\,t-1} + c\sum X_{2,\,t-1}^2. \tag{2A.14c}$$

Substituting $y_t = Y_t - \hat{Y}$, $x_{1,\,t-1} = X_{1,\,t-1} - \hat{X}$ and $x_{2,\,t-1} = X_{2,\,t-1} - \hat{X}$ for Y_t, $X_{1,\,t-1}$ and $X_{2,\,t-1}$, Eqs. (2A.14a), (2A.14b), and (2A.14c), respectively,

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can be rewritten as

$$0 = \mathrm{na} + \mathrm{b}(0) + \mathrm{c}(0), \tag{2A.15a}$$

$$\sum x_{1,t-1}y_t = a(0) + b\sum x_{1,t-1}^2 + c\sum x_{1,t-1}x_{2,t-1}, \tag{2A.15b}$$

$$\sum x_{2,t-1}x_t = a(0) + b\sum x_{1,t-1}x_{2,t-1} + c\sum x_{2,t-1}^2. \tag{2A.15c}$$

The important conditions used to obtain Eq. (2A.15) are

(i) $\sum y_t = 0$,
(ii) $\sum x_{1,t-1} = 0$,
(iii) $\sum x_{2,t-1} = 0$.

Equation (2A.15a) implies that $a = 0$. Therefore, we can use Eqs. (2A.15b) and (2A.15c) to solve $\hat{b}$ and $\hat{c}$. Based upon Cramer's rule, we have

$$\hat{b} = \frac{\sum x_{1,t-1}y_t(\sum x_{2,t-1}^2) - \sum x_{2,t-1}y_t \sum x_{1,t-1}x_{2,t-1}}{(\sum x_{1,t-1}^2)(\sum x_{2,t-1}^2) - (\sum x_{1,t-1}x_{2,t-1})^2}, \tag{2A.16a}$$

$$\hat{c} = \frac{\sum x_{2,t-1}y_t(\sum x_{1,t-1}^2) - \sum x_{1,t-1}y_t \sum x_{1,t-1}x_{2,t-1}}{(\sum x_{1,t-1}^2)(\sum x_{2,t-1}^2) - (\sum x_{1,t-1}x_{2,t-1})^2}. \tag{2A.16b}$$

Substituting Eqs. (2A.16a) and (2A.16b) into Eq. (2A.14a), and dividing both sides of Eqs. (2A.14a) by n, we have

$$\hat{a} = \hat{Y} - \hat{b}\overline{X}_1 - \hat{c}\overline{X}_2. \tag{2A.17}$$

Using the concept of estimating standard errors of regression parameters as discussed earlier, standard errors of $\hat{a}$, $\hat{b}$ and $\hat{c}$ (S_a, S_b, and S_c, respectively) can be estimated. The empirical results for JNJ's ratio are listed in Eq. (2A.13b).

The multiple regression result associated with JNJ's dynamic CR adjustment process is

$$\underset{\quad}{Y_t} = \underset{(0.4323)}{-0.2837} + \underset{(0.3288)}{0.7564}X_{1,t-1} + \underset{(0.2240)}{0.2990}X_{2,t-1}. \tag{2A.13b}$$

Figures below the regression coefficients are standard errors of estimated regression coefficients.

Substituting related estimates into (2A.17), we obtain

$$\begin{aligned}\hat{a} &= 1.7071(0.7564)(1.8448)\ (0.2990)(1.6904)\\ &= 0.2837.\end{aligned}$$

There exists a similar coefficient of determination (R^2) for the multiple regression.

The difference between $Y\frac{1}{2}t$ and its mean, $Y^{\text{TM}\sim}$, can be broken down as

$$(Y_t - \overline{Y}_t) = (Y_t - \hat{Y}_t) + (\hat{Y}_t - \overline{Y}_t), \tag{2A.18}$$

where

$$\hat{Y}_t = \hat{a} + \hat{b}X_{1,\,t-1} + \hat{c}X_{2,\,t-1}. \tag{2A.19}$$

Note that $\hat{Y}$ is the estimated dependent variable and Y_t is the actual dependent variable; $Y_t - \hat{Y}$ is the residual term for the tth period, and $(\hat{Y}_t - \overline{Y})$ represents the difference between the overall mean and the estimated dependent variable in period t (conditional mean). Squaring both sides of Eq. (2A.18) and summing over all observations (1 to n), we obtain

$$\underset{\text{TSS}}{\sum(Y_t - \overline{Y}_t)^2} = \underset{\text{ESS}}{\sum(Y_t - \hat{Y}_t)^2} + \underset{\text{RSS}}{\sum(\hat{Y}_t - \overline{Y}_t)^2}, \tag{2A.20}$$

where

TSS = Total sum of squares;
ESS = Residual sum of squares; and
RSS = Regression sum of squares.

The R^2 can now be defined as

$$R^2 = \frac{\text{RSS}}{\text{TSS}} = \frac{\sum(\hat{Y}_t - \overline{Y}_t)^2}{\sum(Y_t - \overline{Y}_t)^2} = 1 - \frac{\sum \hat{\varepsilon}_t^2}{\sum(Y_t - \overline{Y}_t)^2}. \tag{2A.21}$$

Here R^2 measures the portion of variation in Y that is explained by multiple regression. The term R^2 is often informally interpreted as a measure of "goodness-of-fit" and used as a statistic for comparison of the validity of the regression results under alternative specifications of the independent variables of the model.

The difficulty with R^2 as a measure of "goodness-of-fit" is that R^2 pertains to the explained and unexplained variation in Y: therefore, it does not account for the number of degrees of freedom in the problem. Originally there are n degrees of freedom in a sample of n observations, but one degree of freedom is used up in calculating $\overline{Y}$; thus, leaving only $n - 1$ degrees of freedom for residuals $(Y - \overline{Y})$ to calculate $Var(Y_t)$. Similarly, k degrees of freedom are used up in calculating k regressors; thus, leaving only $n - k$

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degrees of freedom for calculating $Var(\hat{\varepsilon}_t)$. To correct this problem, the corrected R^2 ($\overline{R}^2$) is defined as

$$R^2 = 1 - \frac{\sum \hat{\varepsilon}_t^2}{Var(Y_t)}, \tag{2A.22}$$

where

$$Var(\hat{\varepsilon}_t) = \sigma_{\varepsilon}^2 = \frac{\sum \hat{\varepsilon}_t^2}{n-k},$$

$$Var(Y_t) = \frac{\sum (Y_t - \overline{Y})^2}{n-1}.$$

and $k =$ the number of independent variables.

Even though the ESS will decrease (or remain the same) as new explanatory variables are added, the residual variance may not decrease. From Eqs. (2A.21) and (2A.22), the relationship between R^2 and $\overline{R}^2$ can be defined as

$$\overline{R}^2 = 1 - (1 - R^2)\frac{n-1}{n-k}. \tag{2A.23}$$

Some implications of Eq. (2A.23) are

(a) If $k = 1$, then $R^2 = \overline{R}^2$;
(b) If $k > 1$, then $R^2 > \overline{R}^2$;
(c) $\overline{R}^2$ can be negative.

The R^2 and $\overline{R}^2$ associated with Eq. (2A.13b) are 0.4901 and 0.4350, respectively. If the error, $\hat{\varepsilon}_t$, is normally distributed, t-tests can be applied to test the regression coefficient because $(\hat{a} - a)/S_a$, $(\hat{b} - b)/S_b$ and $(\hat{c} - c)/S_c$ are all t-distributions with $n - k$ degrees of freedom. The t-statistics associated with $\hat{a}$, $\hat{b}$, and $\hat{c}$ are -0.6565, 2.3005, and 1.3345, respectively. The estimated regression coefficients, the standard errors and t-statistics, and $\overline{R}^2$ are printed out by most regression programs.

In addition to the t-statistics, the regression program also prints F-statistics, R^2 and $\overline{R}^2$. The relationship between the F-statistic and R^2 can be defined as

$$F(k-1, n-k) = \frac{R^2}{1-R^2}\frac{n-k}{k-1},$$

where $F(k-1, n-k)$ represents F-statistic with $k-1$ and $n-k$ degrees of freedom.

Strictly speaking, the $F(k-1, n-k)$ F-statistic allows us to test the hypothesis that none of the explanatory variables help explain the variation of Y about its mean. In other words, the F-statistic is used to test the joint hypothesis, H (0): $a = b = c = 0$. The F-statistic for JNJ's current ratio is $F(2, 24) = 11.7689$. This statistic is significantly different from zero at the 95% confidence level. The multiple regression explored here will be used in the chapters on cost of capital and dividend policy.

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Appendix 2.B: Using Indirect Method to Compile Cash Flow Statement

It is important to note that the preparation of the cash flows statement is different from the other three basic financial statements — it is not prepared from the adjusted trial balance sheet, since the statement requires information concerning changes in account balances over a period of time. The information used to prepare this statement comes from comparative balance sheets as shown in Table 2.1, the current income statement as shown in Table 2.2) and selected transaction data to determine how the company used cash during the period.

A review of the income statement reveals information that would be used in preparing the SCF. There are two important facts to be noted about the income statement: One is that the company does not report any depreciation on the statement; and Second is that interest revenue is included in operating income. In most cases, companies place interest received in the investing section but interest paid is shown in the operating section. Similarly, most firms place depreciation and gain or loss from sale of assets in the operating section but cash paid for operating assets is shown in the investing section. As shown below in the SCF, JNJ shows both interest received and interest paid as operating activities and shows all asset related transactions, including depreciation, in the investing section.

Table 2B.2 shows the statement of cash flows for JNJ as submitted to the SEC. Note that, like most other submitted statements, J&J also uses the **indirect method**.

Table 2B.1. Comparative Balance Sheets JNJ.

	2012	2011	Change
Cash/Equivalents	14,911,000.00	24,542,000.00	**−9,631,000.00**
Marketable Securities	6,178,000.00	7,719,000.00	−1,541,000.00
Accounts Receivables — Gross	11,309,000.00	10,581,000.00	728,000.00
Doubtful Account	−466,000.00	−361,000.00	−105,000.00
Raw Materials	1,416,000.00	1,206,000.00	−210,000.00
Goods in Process	2,262,000.00	1,637,000.00	625,000.00
Finished Goods	3,817,000.00	3,442,000.00	375,000.00
Deferred Taxes	3,139,000.00	2,556,000.00	583,000.00
Prepaid & Others	3,084,000.00	2,633,000.00	451,000.00
Total Current Assets	**46,116,000.00**	**54,316,000.00**	**-8,200,000.00**
Land Improvement	793,000.00	754,000.00	39,000.00
Buildings	10,046,000.00	9,389,000.00	657,000.00
Machinery & Equipment	21,075,000.00	19,182,000.00	1,893,000.00
Construction	2,740,000.00	2,504,000.00	236,000.00
Depreciation	−18,557,000.00	−17,090,000.00	−1,467,000.00
Goodwill	22,424,000.00	16,138,000.00	6,286,000.00
Intangible — Gross	18,755,000.00	8,716,000.00	10,039,000.00
Amortization of Intangible	−4,030,000.00	−3,432,000.00	−598,000.00
Deferred Taxes	4,541,000.00	6,540,000.00	−1999,000.00
Other Assets	3,417,000.00	3,773,000.00	−356,000.00
Total Assets	**121,347,000.00**	**113,644,000.00**	**7,703,000.00**
Loans and Notes Payable	4,676,000.00	6,658,000.00	−1,982,000.00
Accounts Payable	5,831,000.00	5,725,000.00	106,000.00
Accrued Liabilities	7,299,000.00	4,608,000.00	2691,000.00
Accrued Compensation	2,423,000.00	2,329,000.00	94,000.00
Accrued Rebates	2,969,000.00	2,637,000.00	332,000.00
Income Taxes	1,064,000.00	854,000.00	210,000.00
Total Current Liabilities	**24,262,000.00**	**22,811,000.00**	**1,451,000.00**
Long-term debt	11,489,000.00	12,969,000.00	−1,480,000.00
Total Long Term Debt	11,489,000.00	12,969,000.00	−1,480,000.00
Deferred Taxes	3,136,000.00	1,800,000.00	1,336,000.00
Employee Oblig.	9,082,000.00	8,353,000.00	729,000.00
Other Liabilities	8,552,000.00	10,631,000.00	−2,079,000.00
Total Liabilities	**56,521,000.00**	**56,564,000.00**	**−43,000.00**
Common Stock	3,120,000.00	3,120,000.00	0.00
Retained Earnings	85,992,000.00	81,251,000.00	4,741,000.00
Compreh. Income	−5,810,000.00	−5,632,000.00	178,000.00
Treasury Stock	−18,476,000.00	−21,659,000.00	−3,183,000.00
Total Equity	**64,826,000.00**	**57,080,000.00**	**7,746,000.00**
Total Liabilities & Shareholders' Equity	**121,347,000.00**	**113,644,000.00**	**7,703,000.00**

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Table 2B.2. Statement of Cash Flows (JNJ).

Johnson & Johnson (JNJ) Company
Statement of Cash Flow for the year ended December 30, 2012

Net Income/Starting Line	**10,514,000.00**
Depreciation	3,666,000.00
Depreciation/Depletion	3,666,000.00
Deferred Taxes	−39,000.00
Other Non-Cash Items	754,000.00
Non-Cash Items	754,000.00
Accounts Receivable	−9,000.00
Inventories	−1,000.00
Other Assets	−2,172,000.00
Payable/Accrued	2,768,000.00
Other Liabilities	−2,555,000.00
Changes in Working-Capital	−2,059,000.00
Cash from Operating Activities	**15,396,000.00**
Purchase of Fixed Assets	−2,934,000.00
Capital Expenditures	−2,934,000.00
Acquisition of Business	−4,486,000.00
Sale of Fixed Assets	1,509,000.00
Sale/Maturity of Investment	14,797,000.00
Purchase of Investments	−13,434,000.00
Other Investing Cash Flow	38,000.00
Other Investing Cash Flow Items, Total	−1,576,000.00
Cash from Investing Activities	**−4,510,000.00**
Cash Dividends Paid — Common	−6,614,000.00
Total Cash Dividends Paid	−6,614,000.00
Repurchase/Retirement of Common	−12,919,000.00
Common Stock, Net	−12,919,000.00
Options Exercised	2,720,000.00
Issuance (Retirement) of Stock, Net	−10,199,000.00
Short-Term Debt Issued	3,268,000.00
Short-Term Debt Reduction	−6,175,000.00
Short-Term Debt, Net	−2,907,000.00
Long-Term Debt Issued	45,000.00
Long-Term Debt Reduction	−804,000.00
Long-Term Debt, Net	−759,000.00
Other*	−83,000.00
Issuance (Retirement) of Debt, Net	−3,749,000.00
Cash from Financing Activities	**−20,562,000.00**

(*Continued*)

Table 2B.2. (*Continued*)

Foreign Exchange Effects	45,000.00
Net Change in Cash	**−9,631,000.00**
Net Cash − Beginning Balance	24,542,000.00
Net Cash − Ending Balance	14,911,000.00
Cash Interest Paid	616,000.00
Cash Taxes Paid	2,507,000.00

*The net cash inflow or outflow from other financing activities. This element is used when there is not a more specific and appropriate element in the taxonomy.

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2.B.1. *Cash Flow from Operating Activities*

In the above statement, the company started with the net income, as reported in the income statement, and adjusted it for all:

1. Non-cash revenues and expenses (For example, depreciation expense).
2. Increases (decreases) in the balances of all operating assets are subtracted (added): For example, when accounts receivable increase during the year, it implies that cash receipts were lower than revenues. JNJ's accounts receivable increased by $9,000 during the year. This amount will be deducted from the net income to arrive at the amount of cash from operations. Similarly, increase in inventories, or any other operating asset, represents an operating use of cash, even though it is not an expense. This will also be deducted from net income to arrive at the cash flow from operations.
3. Increases (decreases) in the balances of all operating liability accounts are added (subtracted). For example, when accounts payable increase during the year, it means the company paid less cash than the expenses incurred. JNJ's accounts payable increased by $2,768,000 during the year. This amount will be added back to net income to arrive at the amount of cash from operations. A similar adjustment is made for the increase in deferred taxes or any other expenses payable, which keeps cash in the business.

A correct procedure to arrive at the amount of net cash flow from operations is to measure change in each operating account on the balance sheet to determine whether any changes in balance sheet accounts caused an increase

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or decrease in cash. A few important points should be noted here:

1. Changes in operating accounts shown on JNJ's SCF do not equal the balance sheet changes. For example, the SCF shows that accounts receivables increased by $9,000. However, the balance sheet reports an increase of only $728,000. Similarly, accounts payable increase by $2,768,000 as per the SCF but the balance sheet shows an increase of $106,000 in accounts payable. It is not clear what accounts for these and other discrepancies in operating assets and liabilities. Perhaps the company has made some adjustments and aggregations, but they are not explained.
2. The SCF shows an increase of deferred taxes of $39,000. A careful look at the balance sheet reveals that deferred taxes are reported thrice: in current assets, in long-term assets and also in long-term liabilities. Moreover, the total of the three is not equal to the amount reported in the SCF.
3. JNJ reports an increase in almost all of its long-term assets in the balance sheet, (land, building, machinery and equipment, construction and goodwill all went up significantly) but the SCF provides only a single number of additions to PP&E of $2,934.000, which does not add up to all the increases in assets. Moreover, the SCF reports proceeds from sale of assets in the investing section, but which assets were sold and whether they were sold at a gain or a loss is not revealed. The company also reports significant transactions in investments and in short term debt instruments in the SCF, but the balance sheet does not even have a category named investments in its asset section. Moreover, the short-term debt does not match the short-term loans payable in the balance sheet. Why is this information not deemed important and why the cash paid for operating assets is not shown in the operating section are major concerns that FASB should address.

2.B.2. *Cash Flow from Investing Activities*

Capital expenditures for long-term assets such as plant, property and equipment, land and buildings are the primary component of investing cash flow. Capital expenditures may be reported net or gross of proceeds from the sale of these assets. However, trends in gross capital expenditures contain useful insight into management plans. The JNJ cash flow statement reports only a net figure for all plant property and equipment, without providing details about changes in their composition.

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Other components of cash flow from investing activities include consequences of acquisitions, investments in affiliates and joint ventures. However, segregating operating assets and liabilities acquired in acquisitions may provide useful information about payments made in cash. The JNJ cash flow statement reports a capital expenditure of $2,934,000 for purchase of various fixed assets. Moreover, they report a cash outflow of $4,486,000 for acquisition of business and another cash outflow of $1,576,000 as "Other cash outflow". It is clear that the company engaged in significant acquisition activity during the year. Since most of the company's long-term assets went up during the year, it is impossible to figure out how much of the increase in assets is due to purchase of new assets and how much is due to acquisition of another business and its assets and liabilities. This is probably the reason why the changes in the balance sheet accounts do not correspond to changes in the cash flow statement.

Please note, the information from comparative balance sheets and the current income statement are not enough to compile the cash flow statement as indicated in Table 2B.2. In other words, we need to get information from either foot note or financial statement or information from JNJ's internal financial reports. Therefore, most accounting textbooks use hypothesized data to compile a cash flow statement in terms of indirect method.

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